

Mathematical Stories

Mathematical Stories

Lecture Notes

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Introduction

Purpose and Philosophy

How to read this course

These notes separate the mathematical core from the supporting argument. The boxed definitions, theorems, formulas, examples, and summaries identify what a student should be able to find and retain. The surrounding derivations, proofs, diagrams, and commentary explain why those statements are true and what they mean.

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

Working principle

The course begins with geometric objects and questions, introduces notation only when it is needed, derives formulas from those objects, and then uses examples to test the resulting language. The reader should always distinguish the object, its coordinate description, the argument connecting them, and the final result.

How the Course Works

Two uses of the same chapter

Each chapter supports the live lecture and later independent study. During the lecture it provides the mathematical route and the core results to discuss. After the lecture it remains a complete narrative containing definitions, examples, arguments, and interpretations that the student can revisit without the lecturer being present.

This book is part of a weekly learning process rather than material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. In a fifteen-week semester, the remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

The Weekly Learning Cycle

The complete weekly cycle

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant objects and methods should be read.
2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF contains the full argument, not merely the material shown or discussed in detail live.
3. **Ask questions and clarify the material.** Mark every unclear definition, transition, calculation, or conclusion and resolve it using the text, discussion, examples, and appropriate tools.
4. **Solve the complete assigned problem set.** Move from following explanations to constructing arguments independently.
5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion.
6. **Present and discuss your work.** Be prepared to explain a solution, answer questions about its steps, and describe what was learned while preparing it.

Why the stages form one process

The lecture prepares the reading, the reading prepares the problem solving, the written solution prepares the presentation, and feedback leads back to revision. No stage is intended to replace the others.

Exercise Classes: Presentation, Feedback, and Revision

What a presentation must demonstrate

A student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether it can be communicated clearly. Feedback given during exercise classes is intended for the whole group, not only for the student whose work is being presented. By comparing different solutions, students learn which explanations are complete, which steps need justification, and which weaknesses require revision.

Remark

Feedback concerns the mathematical work and its results, not the personal worth of the student. It should identify what already works and what must be changed to produce a stronger argument.

Revision is part of mathematical work

The first submitted version need not be the final version. Receiving comments, reconsidering choices, correcting errors, and adding missing explanations are normal parts of serious academic and professional work, not evidence of failure.

Progress and Assessment

What progress looks like

By the end of a weekly cycle, the student should be able to identify the main objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Evidence of serious engagement includes:

- regular preparation and completion of assigned problem sets;
- documented attempts rather than bare answers;
- active participation in presentations and discussion;
- separate, self-contained notes for individual problems;
- careful revision after feedback;
- increasing clarity, independence, and mathematical precision.

Remark

A weak first solution or presentation does not determine the final exercise component grade. Sustained work, revision, and demonstrated development can lead to a very good final result. This does not lower the standard; it recognises the process required to reach it.

The examination is assessed separately and tests mathematical knowledge and the ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills needed to complete it successfully.

How to use the course

Attend, reread, question, solve, write, present, and revise. The lecture provides orientation; the chapter preserves the complete argument; independent work turns that argument into understanding.

How to Prepare Mathematical Notes

Purpose of a mathematical note

A mathematical note does not merely record an answer. It shows how the problem is understood, which information matters, why a method is appropriate, how each important step is justified, and how the conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain objects, operations, conditions, and logical relationships. Reading a formula therefore means being able to explain what each object represents, why the expression is meaningful, and what information it provides.

Remark

A sequence of formulas without explanatory sentences rarely communicates a complete argument. Words connecting the formulas are part of the mathematics because they state the logical relationship between successive steps.

Make the Note Self-Contained

The reader should not need another document

Prepare one separate note for each assigned problem, including all subparts. Begin with the problem or a complete and faithful formulation of it. Record what is given, what must be found or proved, and what notation will be used.

Before calculating, identify the mathematical objects involved and decide what would count as a satisfactory answer. After several weeks, opening the note should still reveal both the question and the full solution without searching elsewhere.

Make the Reasoning Visible

Example

Sentences that reveal mathematical reasoning include:

- Because the determinant is nonzero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is nonzero, it determines a line.

- Substituting into the original equation verifies the solution.

Explain important transitions, not every arithmetic move

The amount of explanation should be proportional to the problem. A simple exercise may need only a few sentences; a substantial argument may need many. The goal is not maximal length but complete visibility of every essential step.

Writing principle

Write as briefly as possible, but as completely as necessary.

A Complete Solution

Five visible stages

1. **Understand the problem.** Record the question, the given information, and the required conclusion.
2. **Choose a method.** Select definitions, theorems, or techniques connecting the data with the goal.
3. **Construct the argument.** Carry out the necessary steps and explain why each important transition is valid.
4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning.
5. **State the conclusion.** Answer the original question clearly and in context.

Remark

A bare final answer, an unexplained sequence of formulas, or disconnected calculations constitute incomplete work. Quality cannot be replaced by quantity, but a single polished solution does not replace completing the rest of the assigned problem set.

Working with AI

AI is an assistant, not the author of the student's understanding

AI may clarify notation, generate simpler examples, check reasoning, identify missing steps, and help organise mathematical prose. The student remains responsible for every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Remark

Receiving a polished answer is not the same as understanding it. A note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently.

One-problem-at-a-time workflow

Work on one problem at a time and prepare one complete note for it. Do not paste an entire problem set into one conversation and request all answers at once. Repeat the focused process separately for every problem in the assigned set.

Example

Useful questions to ask while working with AI include:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler explanation of this transition?
- How can the result be checked?
- What should I be able to explain during class?

What this workflow develops

Preparing notes in this way develops the ability to read notation, recognise objects, organise information, construct arguments, and communicate ideas precisely. Repeating the process for the complete problem set builds habits of systematic academic and professional work.

Student GitHub Workspace

The current instructions, problem templates, and technical workflow are maintained in the public repository [dchorazkiewicz/Math_Problems_Repo](#). Students should create a public fork and commit notes to their own fork; the common course repository is a template, not the submission location.

Repository map

- `README.md` explains the workspace and weekly index;
- `SOURCE_MATERIAL.md` maps weeks to the current PDF and LaTeX problem sources;
- `NOTE_GUIDELINES.md` defines a complete note;
- `AI_WORKFLOW.md` describes focused work, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives connected assistants operational rules;
- `docs/` contains weekly folders and one Markdown file per problem.

The workspace uses Markdown, Git and GitHub, MathJax, MkDocs, and GitHub Actions. Because the technical instructions may change independently of this PDF, students should

consult the repository for the current workflow.

Chapter 1

Euclidean Geometry and Cartesian Coordinates

1.1 The Euclidean Geometry Available Before Coordinates

Before introducing coordinates, we recall the geometric world that is already available. At this stage there are no ordered pairs, no axes, and no vectors. There are points, straight lines, segments, circles, and angles, together with relations such as incidence, betweenness, congruence, perpendicularity, and parallelism.

What must be retained

Geometry does not begin with coordinates. Points, lines, circles, right angles, parallelism, congruence, and straightedge-and-compass constructions are already meaningful before any point is assigned a pair of numbers. Coordinates will add an algebraic language to this existing geometry.

Remark

Coordinates will later translate this geometry into arithmetic. They do not create the geometry. We can already construct and compare geometric objects before assigning numbers to all points and angles.

1.1.1 Three Different Levels Must Not Be Confused

The logical hierarchy

1. *Euclid's postulates* describe the geometric world being assumed.
2. *Straightedge-and-compass rules* describe which elementary drawings and intersections may be used in a construction.
3. *Derived constructions* must be built from those elementary operations and justified by a proof.

Remark

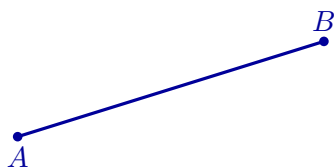
A perpendicular bisector, a perpendicular through a point, an angle bisector, or a parallel through a point is not treated as a primitive permission. Each must be constructed from the elementary toolkit and then justified.

1.1.2 Euclid's Five Postulates

The following are the five postulates used in the first book of Euclid's *Elements*. They specify the elementary geometric operations and relations from which the later constructions begin. The diagrams illustrate their meaning; they are not proofs.

Postulate I: joining two points

A straight segment can be drawn from any point to any other point.



a segment is drawn from A to B

Postulate II: extending a segment

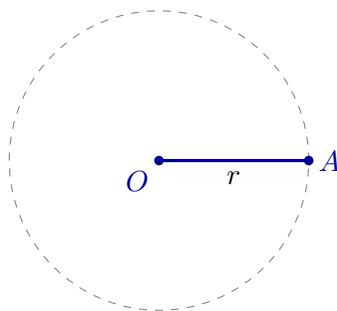
A finite straight segment can be extended continuously in the same straight line.



AB is extended beyond B

Postulate III: drawing a circle

A circle can be drawn with any chosen centre and any chosen radius.



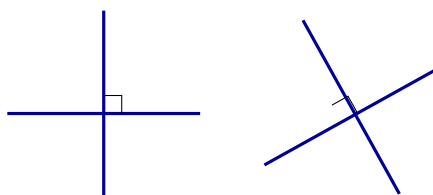
centre O and radius r determine a circle

Why the circle postulate matters

This postulate permits a length to be copied geometrically without first assigning a number to it. The compass transfers a geometric magnitude directly.

Postulate IV: equality of right angles

All right angles are congruent.



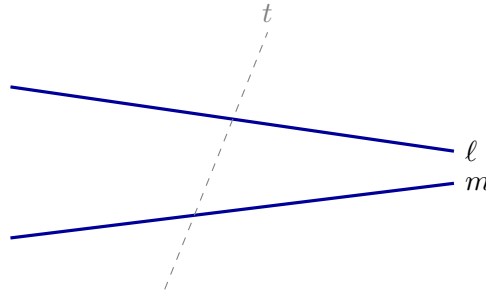
every right angle represents the same geometric magnitude

Remark

This postulate does not say that a perpendicular may simply be drawn without construction. It says that once right angles have been constructed, they are all congruent.

Postulate V: the parallel postulate

If a transversal meets two straight lines and the interior angles on one side sum to less than two right angles, then the two lines, when extended, meet on that side.



the angle condition determines on which side ℓ and m meet

Modern working form

A familiar equivalent is Playfair's formulation: through a point outside a given line there passes exactly one parallel to that line. We shall use that form later, after the relation between the two formulations has been made clear.

What the postulates provide

The first three postulates provide elementary constructions, the fourth makes right angles comparable, and the fifth controls parallelism. Together they specify the geometric world available before coordinates and algebra are introduced.

1.1.3 Elementary Rules for Straightedge-and-Compass Work

For explicit constructions we use the following operational rules.

The elementary construction toolkit

1. Through two already constructed distinct points, draw the straight line.
2. With an already constructed point as centre and an already constructed segment as radius, draw a circle.
3. Retain intersection points of two constructed lines, a line and a circle, or two circles whenever such intersections exist.

How new objects are produced

The first two rules are operational forms of Euclid's first three postulates. The third rule explains how new constructible points arise: they are intersections of objects already obtained. Every later construction in this chapter must be reducible to this toolkit.

1.1.4 Constructions Derived from the Toolkit

The following objects are not added as new primitive permissions. Each is constructed from lines, circles, and their intersections.

The Perpendicular Bisector and the Midpoint

Construction

Let A and B be distinct points. Draw two circles with the same radius, one centred at A and one centred at B , choosing the radius larger than half of $|AB|$. Let their intersection points be P and Q .

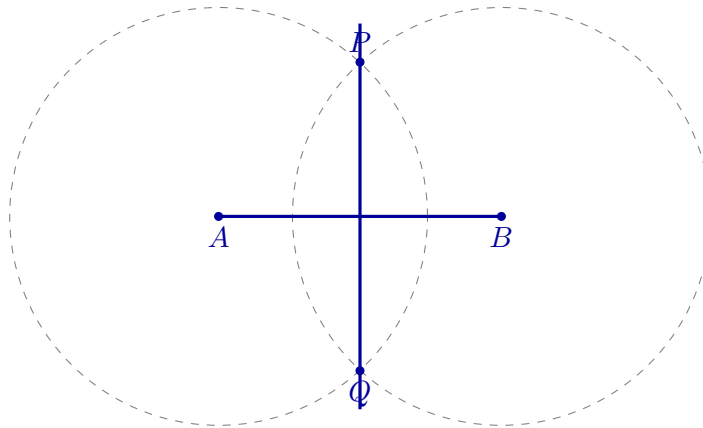
Because P lies on both circles,

$$|PA| = |PB|,$$

and similarly,

$$|QA| = |QB|.$$

Thus both P and Q are equidistant from A and B .



Theorem

The line PQ is the perpendicular bisector of AB . Its intersection with AB is the midpoint of AB . Thus both midpoint and perpendicularity are constructed from equal radii and intersections.

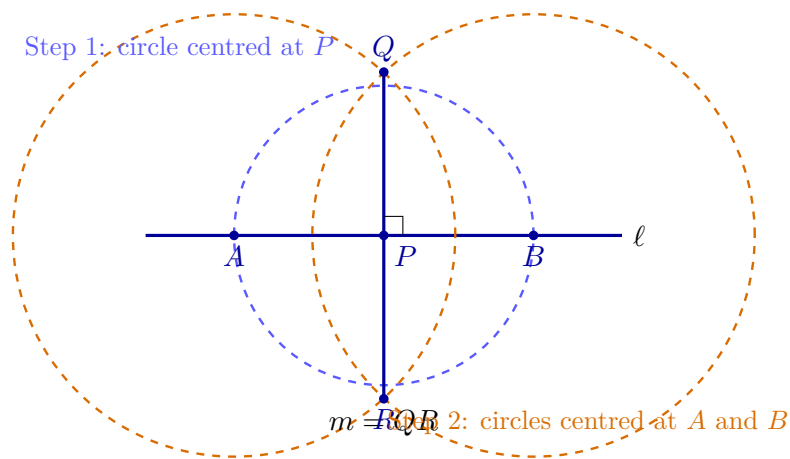
A Perpendicular Through a Point on a Line

Construction

Let P lie on a line ℓ .

Step 1. Draw a circle centred at P . It meets ℓ at two points A and B . Since $|PA| = |PB|$, the point P is the midpoint of AB .

Step 2. Apply the perpendicular-bisector construction to AB . The resulting perpendicular bisector passes through its midpoint P .



Theorem

The constructed line m passes through P and is perpendicular to ℓ .

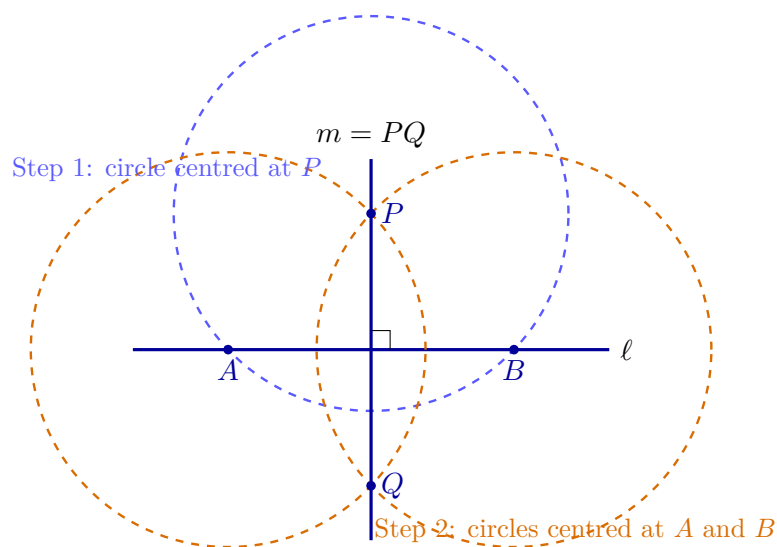
A Perpendicular Through a Point Outside a Line

Construction

Let P lie outside a line ℓ .

Step 1. Draw a circle centred at P with radius large enough to meet ℓ at two points A and B . Then $|PA| = |PB|$, so P lies on the perpendicular bisector of AB .

Step 2. Draw equal-radius circles centred at A and B , chosen so that they meet again at a second point Q . The points P and Q are both equidistant from A and B .



Theorem

The line PQ is the perpendicular bisector of AB and therefore is perpendicular to ℓ through the prescribed external point P .

Bisecting an Angle

Construction and justification

Let two rays with common endpoint O form an angle. Draw a circle centred at O meeting the rays at A and B . Then draw equal-radius circles centred at A and B , and let one of their intersections inside the angle be P . Since

$$|OA| = |OB|, \quad |AP| = |BP|, \quad |OP| = |OP|,$$

the triangles OAP and OBP are congruent.

Theorem

The congruence of OAP and OBP gives

$$\angle AOP = \angle POB.$$

Therefore the ray OP bisects the angle.

Constructing a Parallel Through a Point

Construction

Let P lie outside a line ℓ . First construct a line m through P perpendicular to ℓ . Then construct a line n through P perpendicular to m .

The transversal m forms equal corresponding right angles with n and ℓ . By the converse of the corresponding-angles theorem, the two lines are parallel.

Theorem

The constructed line satisfies

$$n \parallel \ell.$$

The fifth postulate, equivalently Playfair's postulate, guarantees that this is the unique parallel to ℓ through P .

1.1.5 Parallel Lines and Proportional Segments

A central fact of Euclidean geometry is that parallel lines preserve ratios. This is the content of Thales' theorem in the form needed here.

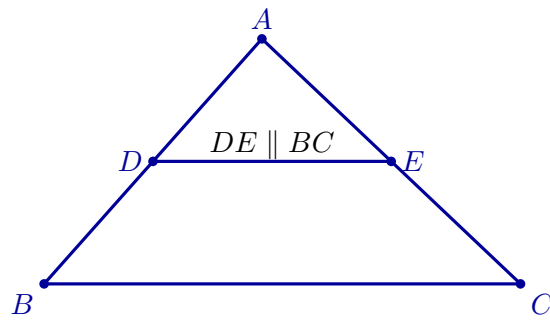
Theorem

Let D lie on AB and E lie on AC in a triangle ABC . If

$$DE \parallel BC,$$

then

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$



Why the ratios are equal

Because $DE \parallel BC$, corresponding angles in triangles ADE and ABC are equal. The triangles therefore have the same three angles and differ only by one common scale factor. Every corresponding side is multiplied by that same factor, which gives the three equal ratios.

Definition

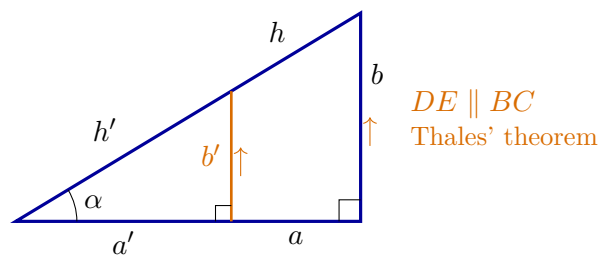
Two triangles are *similar* when their corresponding angles are equal. In that case their corresponding side lengths are proportional.

What similarity preserves

Similarity changes the size of a figure while preserving its shape. It is the reason ratios of corresponding lengths can depend only on angles and later makes trigonometric ratios well defined.

1.1.6 Sine and Cosine Before Coordinates

Take a right triangle containing an acute angle α . Let h denote the hypotenuse, let a be the leg adjacent to α , and let b be the leg opposite α .



$$\frac{a'}{a} = \frac{b'}{b} = \frac{h'}{h}, \quad \frac{a'}{h'} = \frac{a}{h}, \quad \frac{b'}{h'} = \frac{b}{h}.$$

Why the ratios depend only on the angle

All right triangles containing the same acute angle are similar. Corresponding side lengths therefore differ by one common scale factor, which cancels in the ratios a/h and b/h . These ratios depend on the angle, not on the size of the triangle chosen to represent it.

Definition

For an acute geometric angle α ,

$$\cos \alpha = \frac{a}{h}, \quad \sin \alpha = \frac{b}{h}.$$

Definition

For a right angle,

$$\cos \alpha = 0.$$

If α is obtuse and β is its acute supplement, define

$$\cos \alpha = -\cos \beta.$$

Thus cosine is positive for acute angles, zero for a right angle, and negative for obtuse angles.

Trigonometry is geometric before it is coordinate-based

Sine and cosine are ratios determined by similarity. Their definitions do not require Cartesian axes or ordered pairs. Coordinates will later provide a new way to calculate and use these same geometric quantities.

The fundamental identity

For an acute angle, the Pythagorean theorem gives

$$a^2 + b^2 = h^2.$$

Dividing by h^2 yields

$$\left(\frac{a}{h}\right)^2 + \left(\frac{b}{h}\right)^2 = 1.$$

Substituting the definitions of cosine and sine gives the identity below.

Theorem

For every angle in the geometric range considered here,

$$\cos^2 \alpha + \sin^2 \alpha = 1.$$

For acute angles this is a direct reformulation of the Pythagorean theorem.

1.1.7 Angles and Their Numerical Measure

Angle and angle measure are different objects

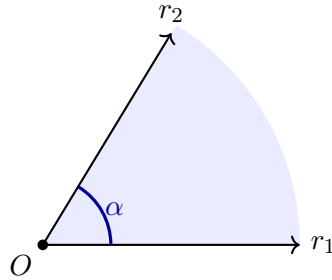
An angle exists geometrically before it is assigned a number. We can compare, copy, add, and bisect angles without degrees or radians. A numerical angle measure requires an additional choice of unit.

Definition

Let r_1 and r_2 be two rays with the same initial point O . The rays, together with a choice of one of the two regions between them, determine an angle α with vertex O .

Remark

The angle is the chosen geometric region. It is neither the small arc drawn near the vertex nor a number. The arc is only a graphical mark indicating which region has been selected.



Why circles give a consistent measure

Draw a circle centred at O . The rays cut from the circle an arc corresponding to the chosen angle.

Theorem

For every Euclidean circle, the ratio of circumference C to diameter D is the same constant:

$$\frac{C}{D} = \pi.$$

Since $D = 2r$, every circle of radius r has circumference

$$C = 2\pi r.$$

Why the quotient s/r is independent of the circle

Similarity implies that enlarging a circle by a factor λ multiplies its diameter, circumference, and corresponding arc lengths by λ . If the same rays cut arc lengths s_1, s_2 from circles of radii r_1, r_2 , then

$$\frac{s_1}{s_2} = \frac{r_1}{r_2}, \quad \text{hence} \quad \frac{s_1}{r_1} = \frac{s_2}{r_2}.$$

The quotient therefore depends only on the angle.

Definition

Let s be the length of the selected arc on a circle of radius r centred at the vertex. The radian measure of α is

$$m_{\text{rad}}(\alpha) = \frac{s}{r}.$$

Full turn, half-turn, and right angle

For a full turn, $s = 2\pi r$, so

$$m_{\text{rad}}(\text{full turn}) = 2\pi.$$

A half-turn therefore has measure π , and a quarter-turn has measure $\pi/2$.

Definition

Degrees provide a second numerical scale. A full turn is assigned 360° , so

$$2\pi \text{ radians} = 360^\circ, \quad m_{\text{deg}}(\alpha) = \frac{180^\circ}{\pi} m_{\text{rad}}(\alpha).$$

Example

A constructible right angle has the two numerical descriptions

$$m_{\text{rad}}(\text{right angle}) = \frac{\pi}{2}, \quad m_{\text{deg}}(\text{right angle}) = 90^\circ.$$

The geometric angle is the same object; only the chosen numerical scale changes.

Sine and cosine as functions of a number

The trigonometric ratios were first defined for acute geometric angles. We now extend them to a full turn. For $t \in [0, 2\pi)$, let $\beta \in [0, \pi/2]$ be the acute reference angle between the direction of t and the nearest horizontal ray. The magnitudes $\sin \beta$ and $\cos \beta$ come from the right-triangle definition; their signs are determined by the quadrant:

	$\cos t$	$\sin t$
$0 \leq t \leq \pi/2$	$+\cos \beta$	$+\sin \beta$
$\pi/2 \leq t \leq \pi$	$-\cos \beta$	$+\sin \beta$
$\pi \leq t \leq 3\pi/2$	$-\cos \beta$	$-\sin \beta$
$3\pi/2 \leq t < 2\pi$	$+\cos \beta$	$-\sin \beta$

Definition

For arbitrary real t , define sine and cosine periodically:

$$\sin(t + 2\pi k) = \sin t, \quad \cos(t + 2\pi k) = \cos t, \quad k \in \mathbb{Z}.$$

From a geometric angle to a numerical input

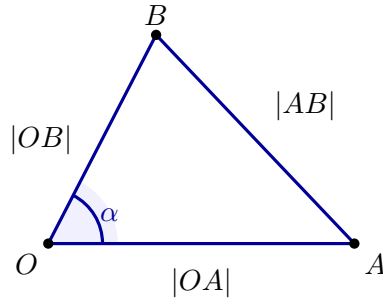
If $t = m_{\text{rad}}(\alpha)$, then $\cos t$ is the same value as the geometric cosine $\cos \alpha$. The notation $\cos t$ therefore treats the numerical measure as an input while preserving the earlier geometric meaning.

The Euclidean law of cosines

Directly measuring an arc is often difficult. A triangle provides another way to recover the measure of an angle. Choose points

$$A \in r_1 \setminus \{O\}, \quad B \in r_2 \setminus \{O\}.$$

The points form the Euclidean triangle OAB .



Theorem

If $\alpha = \angle AOB$, then

$$|AB|^2 = |OA|^2 + |OB|^2 - 2|OA||OB|\cos\alpha.$$

Equivalently, the last factor may be written as $\cos(m_{\text{rad}}(\alpha))$.

Recovering the angle from side lengths

Solving the law of cosines for the angle measure gives

$$m_{\text{rad}}(\alpha) = \arccos\left(\frac{|OA|^2 + |OB|^2 - |AB|^2}{2|OA||OB|}\right).$$

Later, coordinates will provide an arithmetic method for calculating the three Euclidean lengths in this formula.

Remark

At this point no vector has been introduced. We have only Euclidean objects, postulates, constructions, similarity, trigonometric ratios, numerical angle measure, and the Euclidean law of cosines. Vectors will appear only after the Cartesian coordinate system has been constructed.

What numerical measurement adds

The angle is geometric; radians and degrees are numerical descriptions of it. Similarity makes the radian quotient s/r independent of scale, while the law of cosines allows the same angle to be recovered from three side lengths.

Euclidean starting point

Before coordinates we may already construct and compare points, lines, segments, circles, right angles, parallel lines, lengths, and proportions. These geometric objects and relations are the input to the Cartesian construction; they are not consequences of coordinate formulas.

1.2 Constructing Cartesian Coordinates from Euclidean Geometry

The preceding section established the Euclidean objects and constructions available before coordinates. We now make the additional choices that turn that plane into a Cartesian coordinate

plane: axes, an origin, directions, and a common numerical scale.

The central invention

A Cartesian coordinate system converts geometric position into arithmetic data. After choosing two perpendicular axes, an origin, orientations, and a common unit, every point can be described by an ordered pair of real numbers. Geometry can then be studied with algebra without ceasing to be geometry.

1.2.1 The Plane Before Coordinates

Starting point

At first there is only a Euclidean plane. No line has been selected as an axis, no point has been selected as an origin, no orientation has been declared positive, and no numerical scale has been imposed.



a Euclidean plane with no chosen coordinates

Coordinates are additional structure

The Cartesian plane will be obtained by making a sequence of choices inside this pre-existing geometry. Those choices do not change the points or lines; they create a systematic numerical language for describing them.

1.2.2 First Construction: A Straight Line

First choice

Choose two distinct points of the plane and construct the unique straight line through them.

What has and has not been created

At this stage we have only a chosen Euclidean line. It is not yet a coordinate axis: there is no second line, no origin, no positive direction, and no numerical scale.



the first constructed line

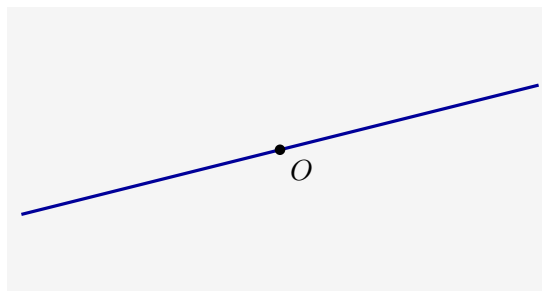
1.2.3 Second Construction: A Point on the Line

Second choice

Choose an arbitrary point O on the constructed line.

A chosen point is not yet an origin

The point O will later become the common zero of both coordinate axes, but it becomes an origin only after the second axis, orientations, and numerical scale have been constructed.



a point chosen on the first line

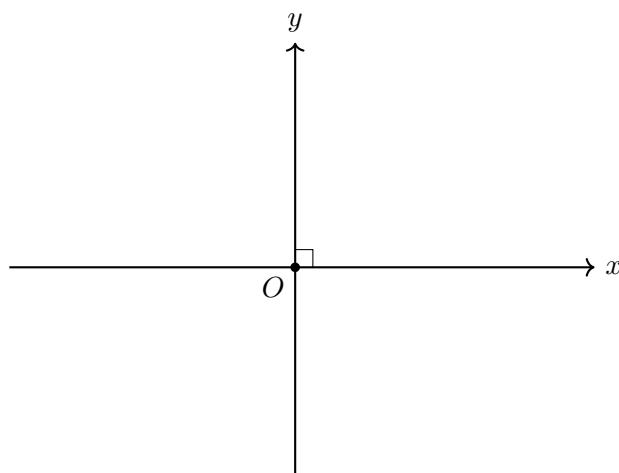
1.2.4 Third Construction: The Perpendicular Line

Third construction

Through O , construct the unique line perpendicular to the first line. The two constructed lines now intersect at a right angle.

Definition

We call the first line the x -axis, the perpendicular line the y -axis, and their intersection O the origin.



What has now been created

Only after the perpendicular line has been constructed do we have the geometric framework of two coordinate axes with a common origin.

Remark

The axes are perpendicular because this was required in their Euclidean construction. No coordinate formula and no numerical angle measure was used to create that right angle.

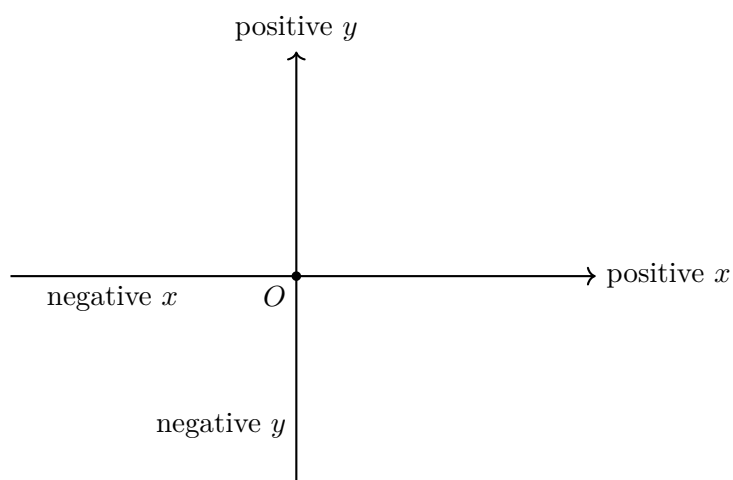
1.2.5 Choosing Positive and Negative Directions

Choosing orientations

Each axis is divided by O into two opposite rays. Choose one ray on the x -axis as positive and one ray on the y -axis as positive. The opposite rays are then called negative.

Orientation is a choice

The Euclidean plane does not force which ray should be positive. Reversing one orientation produces a different coordinate system on the same geometric plane.



What signs will mean

A positive or negative coordinate will record on which oriented side of the origin the corresponding projection lies. The sign is not a geometric length; it is directional information supplied by the chosen orientation.

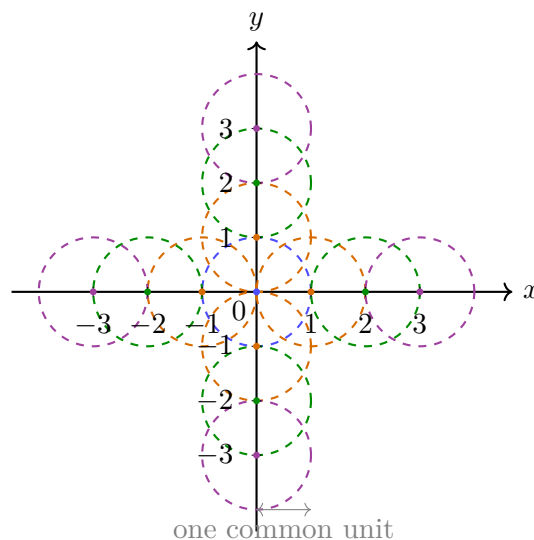
1.2.6 Constructing a Common Unit Length

Choosing and transferring the unit

Choose a positive segment length and call it one unit. Using a compass, transfer this same length from O repeatedly along both axes and in both directions. The first marks are labelled 1 and -1 ; repetition constructs the integer marks

$$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

on each axis.



Why the unit must be common

A segment labelled as one unit on the x -axis must be congruent to a segment labelled as one unit on the y -axis. Otherwise the two coordinate numbers would measure lengths in incompatible scales.

Connection with distance

The common scale is what later allows horizontal and vertical coordinate differences to be combined by the Pythagorean theorem in the standard Euclidean distance formula.

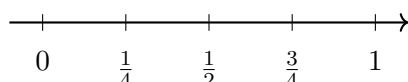
1.2.7 Refining the Scale by Bisection

Constructing finer marks

The compass-and-straightedge construction does not stop at integer marks. Bisecting each unit segment constructs halves. Bisecting again constructs quarters, then eighths, and so on. Repeating this process on every integer interval and on both sides of the origin gives all dyadic marks

$$\frac{k}{2^n}, \quad k \in \mathbb{Z}, \quad n \in \mathbb{N}.$$

successive bisections of one unit



Theorem

The dyadic marks are dense on the axis: every nonempty interval contains a number of the form $k/2^n$. They provide arbitrarily fine numerical approximations to positions on the line.

Definition

Continuity assumption. Once an origin, a positive direction, and a unit length are fixed on a Euclidean line, every point of the line corresponds to exactly one real number, and every real number corresponds to exactly one point of the line.

What continuity contributes

Finite straightedge-and-compass operations construct many particular positions, including every dyadic mark, but not every real position. Continuity supplies the complete real scale. After this additional assumption, every point of each axis has a unique signed real label.

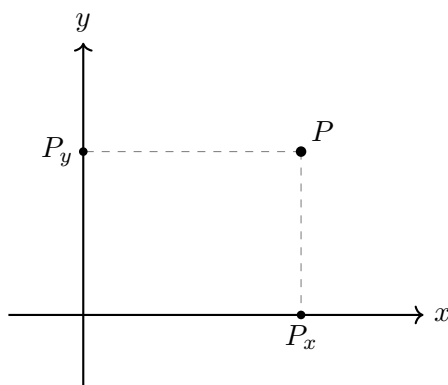
1.2.8 Assigning Coordinates by Orthogonal Projection

When coordinates can be assigned

Only after the axes, orientations, common unit, refined scale, and real labels have been constructed can an arbitrary point of the plane receive a coordinate pair.

Constructing the two coordinate values

Let P be any point of the plane. Through P , construct the line perpendicular to the x -axis; it meets that axis at a unique point P_x . Through P , construct the line perpendicular to the y -axis; it meets that axis at a unique point P_y .



Definition

If the signed real labels of P_x and P_y are x and y , respectively, the coordinates of P are the ordered pair

$$P \longleftrightarrow (x, y).$$

Why the correspondence is one-to-one

The construction is unambiguous because the perpendicular through a given point to a given line is unique. Conversely, given real numbers x and y , locate the corresponding points on the axes and construct through them lines perpendicular to the respective axes. The two lines meet in exactly one point.

Theorem

Once the two perpendicular axes, their positive directions, and a common unit are fixed, orthogonal projection gives a one-to-one correspondence

$$\text{points of the Euclidean plane} \longleftrightarrow \text{pairs } (x, y) \in \mathbb{R}^2.$$

Coordinates are addresses

The ordered pair is the numerical address of a geometric point in the chosen coordinate system. It is not the point itself and it is not the source of the geometry.

1.2.9 The Coordinate Formula for Euclidean Distance

The Euclidean distance between two points is the length of the straight segment joining them. Coordinates now allow this geometric length to be calculated.

The question

Given the Cartesian coordinates of two points, how can the already existing Euclidean length of the segment between them be recovered algebraically?

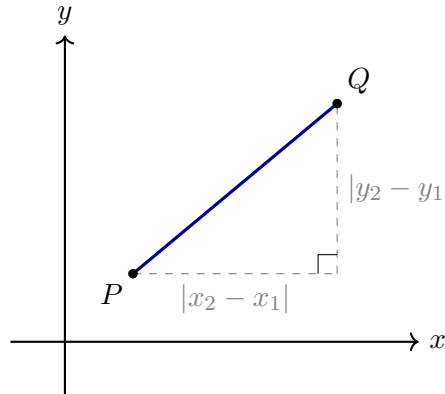
For points

$$P = (x_1, y_1), \quad Q = (x_2, y_2),$$

their projections show that the horizontal and vertical separations have lengths

$$|x_2 - x_1| \quad \text{and} \quad |y_2 - y_1|.$$

Together with the segment PQ , these two perpendicular segments form a right triangle. Here “right” refers to the Euclidean right angle already used in the construction; no numerical measure of that angle is required.



From the right triangle to the formula

By the Pythagorean theorem,

$$d_E(P, Q)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

Distance is nonnegative, so taking the nonnegative square root gives the desired coordinate expression.

Theorem

For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ in the Cartesian plane,

$$d_E(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

What the formula means

The formula does not create a new notion of distance. It is the numerical coordinate expression of the Euclidean segment length already present in the plane. This is the first major payoff of the Cartesian construction: a geometric magnitude has become computable by algebra.

From Euclidean plane to Cartesian distance

The construction proceeds in the following order:

1. begin with a Euclidean plane containing no chosen coordinates;
2. construct one straight line;
3. choose a point O on that line;
4. construct through O the perpendicular line, and only then name the two lines as coordinate axes and O as their origin;
5. choose positive directions on both axes;
6. choose one common unit length and transfer it with a compass in both positive and negative directions to construct the integer marks;

7. repeatedly bisect intervals to construct the dense family of dyadic marks;
8. use Euclidean continuity to identify every axis position with a real number;
9. assign to every point its unique pair of orthogonal projections;
10. derive the coordinate distance formula from the resulting right triangle and the Pythagorean theorem.

Cartesian space

The Cartesian plane is produced by a sequence of explicit choices: two perpendicular axes, their intersection as origin, positive orientations, and a common unit. Orthogonal projection then assigns an ordered pair (x, y) to every point. The distance formula is the algebraic expression of the earlier Euclidean notion of length.

Chapter 2

Vectors, Bases, and Coordinate Systems

2.1 Vectors from Coordinate Displacements

A coordinate pair tells us where a point is. It does not yet tell us how one point is related to another. To describe motion from one point to another we need a new object: not another position, but a record of the change of position.

Choose

$$A = (1, 2), \quad B = (4, 6).$$

To travel from A to B , the horizontal coordinate must increase by 3 and the vertical coordinate by 4. The pair

$$(3, 4)$$

therefore records an instruction:

move three units in the positive x direction and four units in the positive y direction.

This instruction can be carried out from A , but it can also be carried out from any other point. The starting point may change while the displacement remains the same. That distinction is the idea from which vectors arise.

What a vector is

A point records a position. A vector records a displacement: a change of position that can be translated to a different starting point without changing the displacement itself. Vector operations describe how such displacements are combined, reversed, and rescaled.

The development below follows a fixed question-driven order. We first decide when two arrows represent the same displacement. We then ask how successive displacements combine, what it means to repeat or reverse one, what addition and scaling can build, how to measure length and direction, and how two displacements can be compared geometrically. The coordinate formulas will be consequences of those questions, not isolated rules introduced for memorisation.

2.1.1 A Vector Is a Displacement, Not a Place

Let

$$A = (x_1, y_1), \quad B = (x_2, y_2).$$

The point A is a position and the point B is another position. The arrow from A to B asks a different question.

The question answered by a vector

What change of coordinates carries the starting point to the endpoint?

Why endpoint minus starting point appears

The required changes satisfy

$$x_1 + \Delta x = x_2, \quad y_1 + \Delta y = y_2.$$

Solving gives

$$\Delta x = x_2 - x_1, \quad \Delta y = y_2 - y_1.$$

Therefore

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1).$$

This subtraction is forced by the requirement that the displacement carry A to B .

Definition

A vector in the Cartesian plane is a coordinate displacement

$$\mathbf{v} = (v_1, v_2).$$

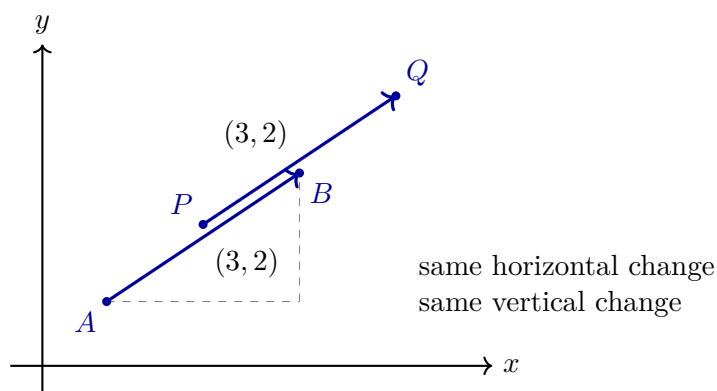
It instructs us to change the first coordinate by v_1 and the second coordinate by v_2 .

A vector is independent of its starting point

If $P = (p_1, p_2)$ and the displacement is $\mathbf{v} = (v_1, v_2)$, then the endpoint is

$$Q = (p_1 + v_1, p_2 + v_2),$$

and $\overrightarrow{PQ} = \mathbf{v}$. Two arrows drawn in different places represent the same vector when they produce the same coordinate changes.



Definition

The zero vector

$$\mathbf{0} = (0, 0)$$

means no displacement. The opposite vector

$$-\mathbf{v} = (-v_1, -v_2)$$

reverses the displacement.

Theorem

If $\mathbf{v}$ carries A to B , then $-\mathbf{v}$ carries B back to A :

$$\overrightarrow{BA} = -\overrightarrow{AB}.$$

2.1.2 Two Journeys Become One: Why Vectors Add

Suppose a point is moved twice. The first displacement is

$$\mathbf{u} = (u_1, u_2),$$

and the second is

$$\mathbf{v} = (v_1, v_2).$$

We want one vector that has exactly the same final effect as these two moves performed successively.

The geometric problem

Vector addition must represent the single displacement whose effect is identical to performing two given displacements one after the other.

Coordinate form forced by successive motion

Start at an arbitrary point (x, y) . After applying $\mathbf{u}$ we are at

$$(x + u_1, y + u_2).$$

Applying $\mathbf{v}$ from there gives

$$(x + u_1 + v_1, y + u_2 + v_2).$$

Relative to the original point, the total horizontal change is $u_1 + v_1$ and the total vertical change is $u_2 + v_2$. Therefore the coordinate rule is forced by the meaning of successive displacement.

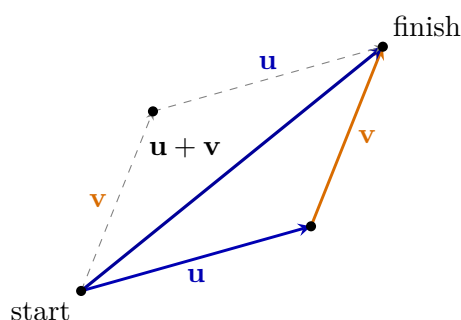
Definition

For vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$, their sum is

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2).$$

What coordinatewise addition records

The formula is bookkeeping for a journey performed in two stages. It adds changes of the same kind: horizontal change to horizontal change and vertical change to vertical change.



Why the order does not change the result

The diagram contains two routes to the same endpoint:

$$\mathbf{u} \text{ then } \mathbf{v}, \quad \mathbf{v} \text{ then } \mathbf{u}.$$

They agree because addition of real numbers is commutative:

$$u_1 + v_1 = v_1 + u_1, \quad u_2 + v_2 = v_2 + u_2.$$

Therefore

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$$

Theorem

Vector addition is commutative:

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$$

Geometrically, this is the parallelogram rule.

Subtraction answers a different practical question: what displacement must be performed after $\mathbf{v}$ in order to have the same effect as $\mathbf{u}$? We seek a vector $\mathbf{w}$ satisfying

$$\mathbf{v} + \mathbf{w} = \mathbf{u}.$$

Subtraction as the missing displacement

Solving the equation $\mathbf{v} + \mathbf{w} = \mathbf{u}$ coordinate by coordinate gives

$$\mathbf{w} = (u_1 - v_1, u_2 - v_2).$$

Definition

For vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$,

$$\mathbf{u} - \mathbf{v} = (u_1 - v_1, u_2 - v_2).$$

In particular,

$$\overrightarrow{AB} = B - A.$$

Remark

The expression $B - A$ does not subtract geometric points as physical objects. It subtracts their coordinate addresses to recover the displacement carrying one point to the other.

2.1.3 Stretching, Shrinking, and Reversing a Displacement

Addition combines different displacements. A second operation appears when the same displacement is repeated.

The geometric idea

Scalar multiplication changes how much of one displacement is taken. It may stretch, shrink, erase, or reverse a vector, but it does not introduce a new line of direction.

From repetition to continuous scaling

If

$$\mathbf{v} = (v_1, v_2),$$

then performing it twice gives

$$\mathbf{v} + \mathbf{v} = (2v_1, 2v_2),$$

and performing it three times gives

$$3\mathbf{v} = (3v_1, 3v_2).$$

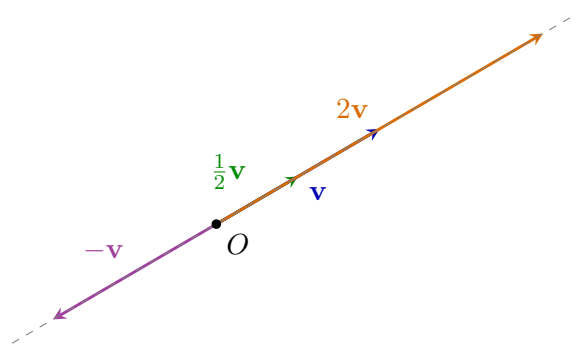
Both components must be multiplied by the same factor if the direction is to remain unchanged. Extending repetition from integers to arbitrary real factors leads to the general operation.

Definition

For a real number λ and a vector $\mathbf{v} = (v_1, v_2)$,

$$\lambda\mathbf{v} = (\lambda v_1, \lambda v_2).$$

The number λ is called a *scalar*.



How the scalar changes the vector

- if $\lambda > 1$, the vector is stretched;
- if $0 < \lambda < 1$, it is shortened;
- if $\lambda = 0$, it collapses to the zero vector;
- if $\lambda < 0$, it is also reversed.

The scalar changes the amount and possibly the orientation of the displacement, not the underlying line on which it lies.

Theorem

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if one is a scalar multiple of the other:

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} = \lambda\mathbf{v} \text{ for some } \lambda \in \mathbb{R} \setminus \{0\}.$$

Remark

Fractions and irrational factors are natural extensions of repetition. The vector $\frac{1}{2}\mathbf{v}$ must be repeated twice to produce $\mathbf{v}$, while $\sqrt{2}\mathbf{v}$ has the same direction and $\sqrt{2}$ times the length. Scalar multiplication turns discrete repetition into continuous scaling.

2.1.4 Linear Combinations: What Addition and Scaling Can Build

Addition and scalar multiplication were introduced as two separate operations, but they are most useful when they are combined.

Definition

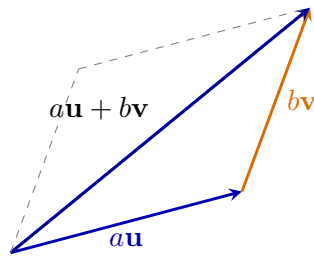
Given vectors $\mathbf{u}$ and $\mathbf{v}$ and real numbers a and b , the vector

$$a\mathbf{u} + b\mathbf{v}$$

is called a *linear combination* of $\mathbf{u}$ and $\mathbf{v}$.

How a linear combination is constructed

First scale $\mathbf{u}$ by a and $\mathbf{v}$ by b ; then add the resulting displacements. A linear combination is therefore not a new operation. It is a controlled use of addition and scalar multiplication.



One direction versus two directions

A single nonzero vector generates only the line

$$\{\lambda\mathbf{v} : \lambda \in \mathbb{R}\}.$$

Two parallel vectors still generate only one line. Two nonparallel vectors provide two independent directions and can generate the entire plane.

Why nonparallel directions reach the plane

If $\mathbf{u}$ and $\mathbf{v}$ are parallel, every scalar multiple of either vector lies on the same line, so every sum $a\mathbf{u} + b\mathbf{v}$ remains on that line. If they are not parallel, the parallelogram construction allows a target displacement to be decomposed into one component along each direction. The two coefficients record those two signed motions.

Theorem

Two nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ determine every vector in the plane uniquely in the form

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$

The numbers a and b record how much motion occurs in each chosen direction.

Bridge to a basis

A basis is not an unrelated new object. It is a pair of independent directions whose linear combinations describe every vector uniquely. Basis coordinates are precisely the coefficients in that reconstruction.

2.1.5 How Much Motion and in Which Direction?

The coordinate pair of a vector records two changes, but it does not immediately answer two basic geometric questions.

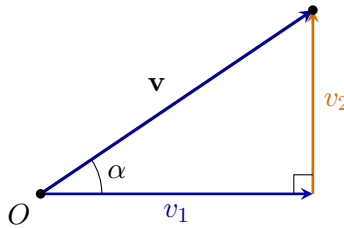
Two questions carried by every nonzero vector

1. How long is the displacement?
2. In which direction does it point?

Let

$$\mathbf{v} = (v_1, v_2)$$

be drawn from the origin. Its endpoint, together with its horizontal and vertical coordinate changes, forms a right triangle. The legs have lengths $|v_1|$ and $|v_2|$, while the vector itself is the hypotenuse.



Length from the Pythagorean theorem

The right triangle gives

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2.$$

The length is nonnegative, so

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

Definition

The Euclidean length, or norm, of $\mathbf{v} = (v_1, v_2)$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

The double bars distinguish the vector from the number measuring its length.

Theorem

$$\|\mathbf{0}\| = 0, \quad \mathbf{v} \neq \mathbf{0} \implies \|\mathbf{v}\| > 0.$$

Moreover, for points A and B ,

$$d_E(A, B) = \|B - A\|.$$

A displacement packages distance and direction

The vector $B - A$ records how to move from A to B , while its norm records the Euclidean distance travelled. The same object therefore carries both directional and metric information.

Now assume $\mathbf{v} \neq \mathbf{0}$ and let α be its angle of inclination, measured from the positive x -axis.

Separating magnitude from direction

The right triangle gives

$$\cos \alpha = \frac{v_1}{\|\mathbf{v}\|}, \quad \sin \alpha = \frac{v_2}{\|\mathbf{v}\|}.$$

Therefore

$$v_1 = \|\mathbf{v}\| \cos \alpha, \quad v_2 = \|\mathbf{v}\| \sin \alpha.$$

Theorem

Every nonzero vector admits the decomposition

$$\mathbf{v} = \|\mathbf{v}\|(\cos \alpha, \sin \alpha).$$

The vector $(\cos \alpha, \sin \alpha)$ has length one.

The two pieces of a vector

- $\|\mathbf{v}\|$ says how much displacement occurs;
- $(\cos \alpha, \sin \alpha)$ says in which direction it occurs.

Thus every nonzero vector is a positive magnitude multiplied by a unit direction.

2.1.6 One Angle Calculation, Two Geometric Tests

Take two arbitrary nonzero vectors

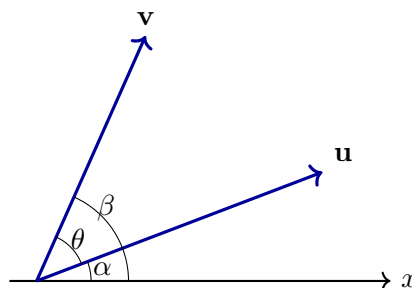
$$\mathbf{u} = (u_1, u_2), \quad \mathbf{v} = (v_1, v_2).$$

Let their angles of inclination be α and β . Then

$$\mathbf{u} = \|\mathbf{u}\|(\cos \alpha, \sin \alpha), \quad \mathbf{v} = \|\mathbf{v}\|(\cos \beta, \sin \beta).$$

One comparison controls both tests

The directed turn from $\mathbf{u}$ to $\mathbf{v}$ is $\beta - \alpha$. Its sine detects parallel directions, while its cosine detects perpendicular directions. The coordinate tests will be derived from these geometric facts.



Parallel directions arise from the sine of the angle difference

Deriving the determinant test

Two nonzero vectors determine the same unoriented direction precisely when

$$\beta - \alpha \equiv 0 \pmod{\pi},$$

equivalently when $\sin(\beta - \alpha) = 0$. Using

$$\sin(\beta - \alpha) = \sin \beta \cos \alpha - \cos \beta \sin \alpha$$

and substituting

$$\cos \alpha = \frac{u_1}{\|\mathbf{u}\|}, \quad \sin \alpha = \frac{u_2}{\|\mathbf{u}\|}, \quad \cos \beta = \frac{v_1}{\|\mathbf{v}\|}, \quad \sin \beta = \frac{v_2}{\|\mathbf{v}\|},$$

we obtain

$$\sin(\beta - \alpha) = \frac{u_1 v_2 - u_2 v_1}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

The denominator is nonzero, so the sine vanishes exactly when the numerator vanishes.

Theorem

For nonzero vectors in the plane,

$$\mathbf{u} \parallel \mathbf{v} \iff u_1 v_2 - u_2 v_1 = 0.$$

Equivalently,

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} = \lambda \mathbf{v} \text{ for some } \lambda \neq 0.$$

Why the determinant condition gives a scalar multiple

If $v_1 \neq 0$, set $\lambda = u_1/v_1$. The equality $u_1 v_2 - u_2 v_1 = 0$ gives $u_2 = \lambda v_2$, hence $\mathbf{u} = \lambda \mathbf{v}$. If $v_1 = 0$, then $v_2 \neq 0$ and the same argument uses $\lambda = u_2/v_2$.

Meaning of the determinant expression

The quantity $u_1 v_2 - u_2 v_1$ measures whether the two coordinate directions span genuine area. It is zero exactly when both vectors lie on one line.

The cosine of the same difference reveals perpendicularity

Deriving the dot-product test

The vectors are perpendicular precisely when

$$\beta - \alpha \equiv \frac{\pi}{2} \pmod{\pi},$$

equivalently when $\cos(\beta - \alpha) = 0$. The identity

$$\cos(\beta - \alpha) = \cos \beta \cos \alpha + \sin \beta \sin \alpha$$

gives

$$\cos(\beta - \alpha) = \frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Since the denominator is nonzero, perpendicularity is equivalent to the vanishing of the numerator.

Definition

For vectors in the plane, the *dot product* is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

In three-dimensional space,

$$(u_1, u_2, u_3) \cdot (v_1, v_2, v_3) = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Theorem

For nonzero vectors,

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

More generally, if θ is the smaller angle between them, then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

What the sign of the dot product says

The dot product is positive for an acute angle, zero for a right angle, and negative for an obtuse angle. It converts angular information into one scalar.

Example

Take

$$\mathbf{u} = (3, 2), \quad \mathbf{v} = (-2, 3).$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 3(-2) + 2(3) = 0.$$

Therefore $\cos \theta = 0$, so the vectors are perpendicular.

Two coordinate tests from one geometric comparison

The sine of the angle difference produces the parallelism test $u_1 v_2 - u_2 v_1 = 0$. The cosine of the same difference produces the perpendicularity test $\mathbf{u} \cdot \mathbf{v} = 0$. Neither rule is imposed arbitrarily; both are consequences of the geometry of angle.

2.1.7 Why the Dot Product Can Be Expanded

The dot product was discovered from the cosine of the angle between two vectors. Before using it in longer calculations, we must verify how it behaves under addition and scalar multiplication.

Verification from coordinates

Let

$$\mathbf{u} = (u_1, u_2), \quad \mathbf{v} = (v_1, v_2), \quad \mathbf{w} = (w_1, w_2).$$

Then

$$\begin{aligned}(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} &= (u_1 + w_1)v_1 + (u_2 + w_2)v_2 \\&= (u_1v_1 + u_2v_2) + (w_1v_1 + w_2v_2) \\&= \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}.\end{aligned}$$

Thus the dot product distributes over vector addition. Similarly,

$$(\lambda \mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v}).$$

The symmetry of multiplication of real numbers also gives

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}.$$

Finally,

$$\mathbf{u} \cdot \mathbf{u} = u_1^2 + u_2^2 = \|\mathbf{u}\|^2.$$

Theorem

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and a real number λ ,

$$\begin{aligned}(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} &= \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}, \\(\lambda \mathbf{u}) \cdot \mathbf{v} &= \lambda(\mathbf{u} \cdot \mathbf{v}), \\ \mathbf{u} \cdot \mathbf{v} &= \mathbf{v} \cdot \mathbf{u}.\end{aligned}$$

Moreover,

$$\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2.$$

Remark

These properties justify every later expansion involving expressions such as $(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$. They are not formal rules borrowed from elsewhere; they follow from the coordinate definition of the dot product.

2.1.8 The Angle Between Two Vectors

Directed turn versus ordinary angle

The difference of inclination angles $\beta - \alpha$ is a directed turn modulo a full turn. The ordinary angle between two nonzero vectors is the smaller, unoriented separation between their directions.

Definition

For nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, their angle is the unique number

$$\theta \in [0, \pi]$$

satisfying

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Why the interval is $[0, \pi]$

The interval chooses the smaller unoriented angle. Directions with inclinations 10° and 350° are separated by 20° , not 340° .

Theorem

The angle distinguishes the principal relative positions:

$$\theta = 0 \iff \text{same direction}, \quad \theta = \frac{\pi}{2} \iff \text{perpendicular}, \quad \theta = \pi \iff \text{opposite directions}.$$

Remark

The Cauchy–Schwarz inequality guarantees that

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1,$$

so the angle is well defined. The inequality will be justified from the vector geometry developed later.

2.1.9 The Cosine Theorem Emerges Immediately

Draw $\mathbf{u}$ and $\mathbf{v}$ from a common point. The displacement from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$ is

$$\mathbf{u} - \mathbf{v}.$$

Thus these three vectors determine a triangle with side lengths

$$\|\mathbf{u}\|, \quad \|\mathbf{v}\|, \quad \|\mathbf{u} - \mathbf{v}\|.$$

Expanding the third side

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|^2 &= (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) \\ &= \mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \|\mathbf{v}\| \cos \theta. \end{aligned}$$

If

$$a = \|\mathbf{u}\|, \quad b = \|\mathbf{v}\|, \quad c = \|\mathbf{u} - \mathbf{v}\|,$$

then the familiar triangle formula follows.

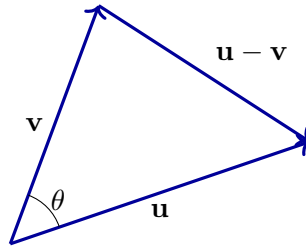
Theorem

For a triangle with side lengths a , b , and c , where θ is the angle between the sides of lengths a and b ,

$$c^2 = a^2 + b^2 - 2ab \cos \theta.$$

Why the theorem appears here

The cosine theorem is not inserted as an external formula. It is the expansion of the squared displacement between the endpoints of two arbitrary vectors. The triangle law is therefore encoded in vector subtraction and the dot product.



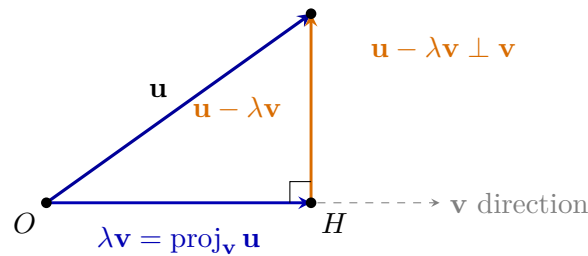
2.1.10 Projection Is Forced by the Right-Angle Condition

Let $\mathbf{v} \neq \mathbf{0}$. We seek a multiple $\lambda \mathbf{v}$ such that the remainder

$$\mathbf{u} - \lambda \mathbf{v}$$

is perpendicular to $\mathbf{v}$. Geometrically, this means decomposing $\mathbf{u}$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$:

$$\mathbf{u} = \lambda \mathbf{v} + (\mathbf{u} - \lambda \mathbf{v}).$$



Geometric requirement

The point H is chosen on the line spanned by $\mathbf{v}$. The vector from O to H is the parallel component $\lambda \mathbf{v}$, while the vector from H to the endpoint of $\mathbf{u}$ is the perpendicular remainder. The right-angle condition determines the value of λ .

Derivation

The perpendicularity criterion requires

$$(\mathbf{u} - \lambda \mathbf{v}) \cdot \mathbf{v} = 0.$$

Expanding gives

$$\mathbf{u} \cdot \mathbf{v} - \lambda(\mathbf{v} \cdot \mathbf{v}) = 0,$$

so

$$\lambda = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}.$$

Definition

For $\mathbf{v} \neq \mathbf{0}$, the projection of $\mathbf{u}$ onto the direction of $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Remark

The formula is the solution of a geometric requirement, not a rule to be memorised without explanation.

2.1.11 Orthogonal Decomposition and Its Consequences

Definition

For $\mathbf{v} \neq \mathbf{0}$, the orthogonal decomposition of $\mathbf{u}$ relative to the direction of $\mathbf{v}$ is

$$\mathbf{u} = \text{proj}_{\mathbf{v}} \mathbf{u} + (\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}).$$

The first term is parallel to $\mathbf{v}$ and the second is perpendicular to $\mathbf{v}$.

Theorem

Because the two components are perpendicular,

$$\|\mathbf{u}\|^2 = \|\text{proj}_{\mathbf{v}} \mathbf{u}\|^2 + \|\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}\|^2.$$

This is the Pythagorean theorem applied to vector components.

Scalar component and vector projection

Definition

The signed scalar component of $\mathbf{u}$ in the direction of nonzero $\mathbf{v}$ is

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \text{comp}_{\mathbf{v}} \mathbf{u} \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Scalar amount and vector displacement

The scalar component measures signed length along the chosen direction. The vector projection turns that signed amount back into an actual displacement by multiplying it by the unit vector $\mathbf{v}/\|\mathbf{v}\|$.

The Cauchy–Schwarz inequality

Projection cannot be longer than the whole vector

The length of the parallel component cannot exceed the length of $\mathbf{u}$:

$$|\text{comp}_{\mathbf{v}} \mathbf{u}| \leq \|\mathbf{u}\|.$$

Substituting the scalar-component formula gives

$$\frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{v}\|} \leq \|\mathbf{u}\|.$$

Multiplying by $\|\mathbf{v}\| > 0$ yields the inequality below.

Theorem

For all vectors $\mathbf{u}$ and $\mathbf{v}$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Consequently,

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1$$

for nonzero vectors, so the angle formula using arccos is meaningful.

The triangle inequality

The direct route is no longer than a broken route

Using Cauchy–Schwarz,

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.\end{aligned}$$

Both sides are nonnegative, so taking square roots gives the desired result.

Theorem

For all vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Geometric meaning

The triangle inequality is the algebraic form of the Euclidean fact that the straight segment between two points is no longer than a route that changes direction at an intermediate point.

2.1.12 From the Plane to Three-Dimensional Vectors

What changes in three dimensions

Adding a third coordinate does not change the meaning of a vector. A vector is still a displacement, now recording three independent coordinate changes:

$$\mathbf{v} = (v_1, v_2, v_3).$$

Addition and scalar multiplication remain componentwise.

Length in three dimensions

Apply the Pythagorean theorem first to the first two coordinate changes and then to the third. This gives

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2 + v_3^2.$$

Definition

For $\mathbf{v} = (v_1, v_2, v_3)$,

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$,

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

Theorem

The geometric role of the dot product is unchanged:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Consequently,

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

A second product available in three dimensions

The new geometric question

Given two vectors $\mathbf{u}$ and $\mathbf{v}$, can we construct a vector perpendicular to both of them?

Definition

For

$$\mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

the *cross product* is

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1).$$

The cross product is perpendicular to both inputs

Direct substitution gives

$$\begin{aligned} (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} &= (u_2v_3 - u_3v_2)u_1 + (u_3v_1 - u_1v_3)u_2 + (u_1v_2 - u_2v_1)u_3 \\ &= 0, \end{aligned}$$

and similarly

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0.$$

Therefore the cross product is perpendicular to both vectors.

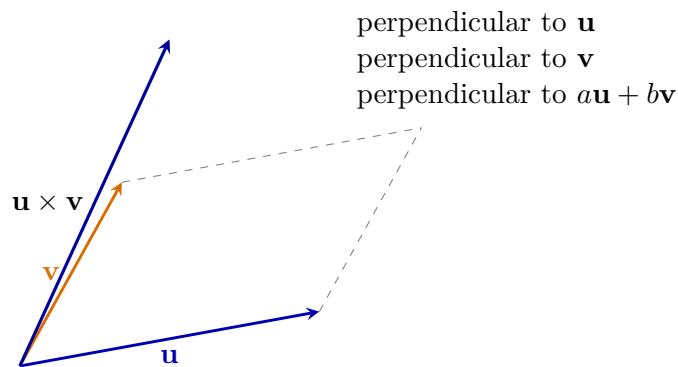
Theorem

$$(\mathbf{u} \times \mathbf{v}) \perp \mathbf{u}, \quad (\mathbf{u} \times \mathbf{v}) \perp \mathbf{v}.$$

Moreover, for all real a, b ,

$$(\mathbf{u} \times \mathbf{v}) \cdot (a\mathbf{u} + b\mathbf{v}) = 0.$$

Thus one cross product is perpendicular to every linear combination of its two inputs.



Remark

We do not yet turn the family $a\mathbf{u} + b\mathbf{v}$ into a theory of planes. That construction belongs to the next chapter. Here the cross product is introduced only as a vector operation and its essential properties are proved.

The cross product detects parallelism

Why zero cross product means parallel directions

If $\mathbf{v} = \lambda\mathbf{u}$, substitution into the coordinate formula makes every component of $\mathbf{u} \times \mathbf{v}$ vanish. Conversely, for nonzero vectors, a zero cross product means that the two direction triples are linearly dependent in no pair of coordinates and therefore lie on one common line.

Theorem

For nonzero vectors,

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}.$$

Together with the dot-product test,

$$\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v},$$

$$\mathbf{u} \times \mathbf{v} = \mathbf{0} \iff \mathbf{u} \parallel \mathbf{v}.$$

Order and orientation

Definition

The direction of $\mathbf{u} \times \mathbf{v}$ is chosen by the right-hand rule: curl the fingers of the right hand from $\mathbf{u}$ toward $\mathbf{v}$; the thumb indicates the positive perpendicular direction.

Theorem

Reversing the order reverses the cross product:

$$\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v}).$$

Its length is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

What the magnitude records

The factor $\sin \theta$ explains the parallelism test: it vanishes for $\theta = 0$ and $\theta = \pi$. The same magnitude will later measure the area of the parallelogram generated by the two vectors.

Dot product and cross product

operation	output	main test
$\mathbf{u} \cdot \mathbf{v}$	number	$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$
$\mathbf{u} \times \mathbf{v}$	vector	$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}$

The cross product is perpendicular to both inputs and to every linear combination of them. These facts will become tools for lines and planes in the following chapter.

Vector toolkit

A vector is a displacement, not a point. The essential operations are addition and scalar multiplication; together they form linear combinations. Length and the dot product encode metric information, while parallelism, perpendicularity, angles, projections, and orthogonal decompositions translate geometric questions into coordinate calculations. The same ideas extend from the plane to three-dimensional space.

2.2 Changing the Basis: New Coordinates for the Same Vector

The preceding section developed the basic geometry and algebra of vectors: addition, scalar multiplication, linear combinations, length, angle, the dot product, parallelism, perpendicularity, and orthogonal projection. We now use that toolkit to study how the coordinate description of one fixed geometric vector changes when the basis is changed.

What changes and what does not

Changing the basis does not change the geometric vector. It changes only the pair of coefficients used to reconstruct that vector from the chosen basis directions. Coordinates belong to a vector relative to a basis; the vector itself is independent of that description.

2.2.1 A Basis Is a Pair of Independent Directions

Up to this point every vector has been described with respect to the horizontal and vertical coordinate directions. That choice was natural because the Cartesian axes had already been constructed, but it was still only a choice. The geometric vector itself does not contain a preferred pair of coordinate axes.

The question of basis coordinates

What numbers describe the same vector when we choose two different coordinate directions? The vector remains fixed. Only the language used to describe it changes.

Definition

Two nonparallel vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ form a basis of the plane. Every vector $\mathbf{u}$ can be written in one and only one way as

$$\mathbf{u} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2.$$

The numbers c_1 and c_2 are the coordinates of $\mathbf{u}$ in the ordered basis

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2), \quad [\mathbf{u}]_{\mathcal{B}} = (c_1, c_2).$$

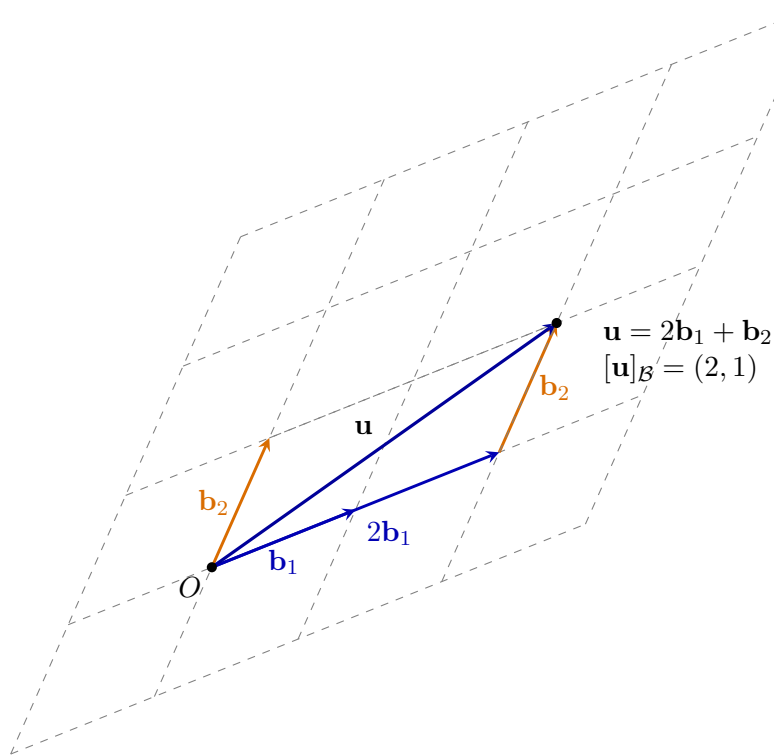
The basis vectors need not be perpendicular and need not have length one. They must only be nonparallel. Otherwise both directions collapse onto one line and cannot describe every displacement in the plane.

Coordinates as signed instructions

The equation

$$\mathbf{u} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2$$

means: move by $c_1 \mathbf{b}_1$, then by $c_2 \mathbf{b}_2$. The endpoint must coincide with the endpoint of $\mathbf{u}$. The coefficients are signed amounts of the chosen basis directions, not lengths of perpendicular shadows.



2.2.2 The Same Vector on Two Different Grids

Example

Let the ordinary Cartesian basis be

$$\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2), \quad \mathbf{e}_1 = (1, 0), \quad \mathbf{e}_2 = (0, 1),$$

and choose the slanted basis

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2), \quad \mathbf{b}_1 = (2, 1), \quad \mathbf{b}_2 = (1, 2).$$

Consider the geometric vector

$$\mathbf{u} = (5, 4)$$

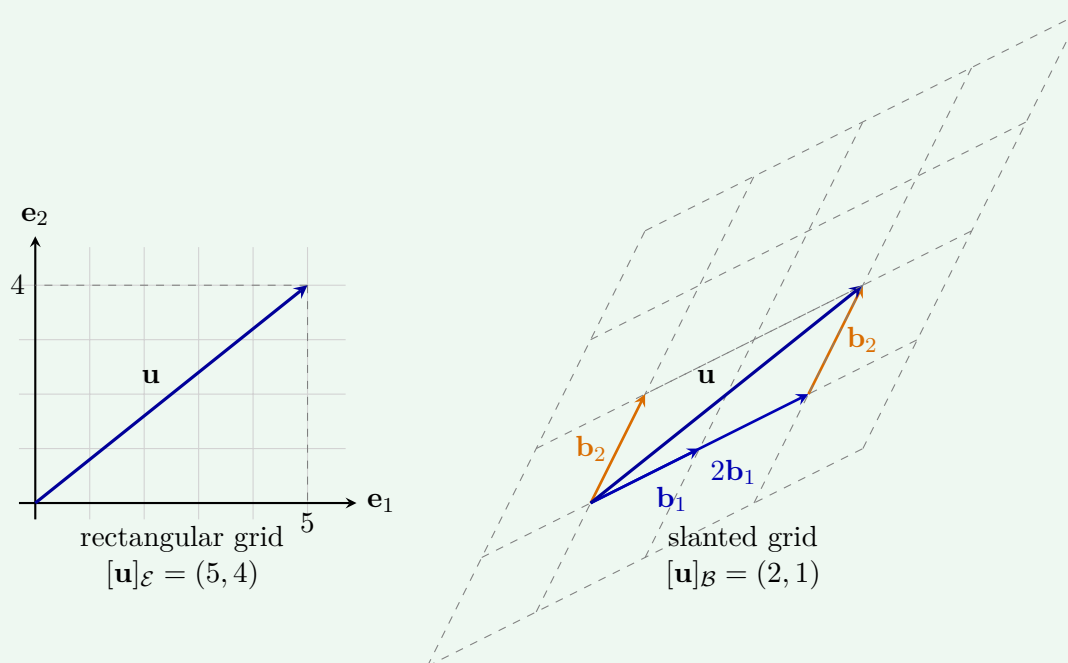
in Cartesian coordinates. On the rectangular grid,

$$[\mathbf{u}]_{\mathcal{E}} = (5, 4).$$

On the slanted grid, we seek c_1, c_2 such that

$$\mathbf{u} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2.$$

The two pictures show the same arrow. Only the grid used to read its coordinates has changed.



On the rectangular grid the coordinates are read along perpendicular axes. On the slanted grid the lines through the endpoint are drawn parallel to the other basis vector, producing an oblique parallelogram rather than a rectangle.

2.2.3 Finding New Coordinates from a Linear System

The geometric picture tells us what must be true. To calculate the coefficients, write

$$\mathbf{b}_1 = (a_1, a_2), \quad \mathbf{b}_2 = (b_1, b_2), \quad \mathbf{u} = (x, y).$$

From a vector equation to scalar equations

The equation

$$\mathbf{u} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2$$

means equality of corresponding Cartesian components. Therefore

$$a_1 c_1 + b_1 c_2 = x,$$

$$a_2 c_1 + b_2 c_2 = y.$$

This system of two linear equations is the algebraic content of changing from the Cartesian basis to $\mathcal{B}$.

Example

For

$$\mathbf{b}_1 = (2, 1), \quad \mathbf{b}_2 = (1, 2), \quad \mathbf{u} = (5, 4),$$

we solve

$$2c_1 + c_2 = 5,$$

$$c_1 + 2c_2 = 4.$$

From the first equation,

$$c_2 = 5 - 2c_1.$$

Substituting into the second gives

$$c_1 + 2(5 - 2c_1) = 4,$$

so

$$-3c_1 = -6, \quad c_1 = 2.$$

Then

$$c_2 = 5 - 2 \cdot 2 = 1.$$

Hence

$$[\mathbf{u}]_{\mathcal{B}} = (2, 1).$$

The calculation agrees with the oblique parallelogram.

2.2.4 A General Coordinate Formula in the Plane

Eliminating the basis coefficients

Starting from

$$a_1c_1 + b_1c_2 = x,$$

$$a_2c_1 + b_2c_2 = y,$$

multiply the first equation by b_2 and the second by b_1 :

$$a_1b_2c_1 + b_1b_2c_2 = xb_2,$$

$$a_2b_1c_1 + b_1b_2c_2 = yb_1.$$

Subtracting eliminates c_2 :

$$(a_1b_2 - a_2b_1)c_1 = xb_2 - yb_1.$$

Similarly, eliminating c_1 gives

$$(a_1b_2 - a_2b_1)c_2 = a_1y - a_2x.$$

Coordinates in a planar basis

If

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2), \quad \mathbf{b}_1 = (a_1, a_2), \quad \mathbf{b}_2 = (b_1, b_2)$$

is a basis and $\mathbf{u} = (x, y)$, then

$$c_1 = \frac{xb_2 - yb_1}{a_1b_2 - a_2b_1}, \quad c_2 = \frac{a_1y - a_2x}{a_1b_2 - a_2b_1}.$$

Why the denominator matters

The denominator

$$a_1b_2 - a_2b_1$$

is nonzero precisely because the basis vectors are not parallel. If it vanished, the two chosen directions would collapse onto one line and the coordinates would not be uniquely determined.

2.2.5 Changing Directly from One Basis to Another

Suppose

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2) \quad \text{and} \quad \mathcal{C} = (\mathbf{c}_1, \mathbf{c}_2)$$

are two bases and

$$[\mathbf{u}]_{\mathcal{B}} = (p, q).$$

Reconstruct first, then describe again

First reconstruct the geometric vector:

$$\mathbf{u} = p\mathbf{b}_1 + q\mathbf{b}_2.$$

If

$$\mathbf{b}_1 = (a_1, a_2), \quad \mathbf{b}_2 = (b_1, b_2),$$

then

$$\mathbf{u} = (pa_1 + qb_1, pa_2 + qb_2).$$

Now seek r, s such that

$$\mathbf{u} = r\mathbf{c}_1 + s\mathbf{c}_2.$$

Writing

$$\mathbf{c}_1 = (d_1, d_2), \quad \mathbf{c}_2 = (e_1, e_2),$$

we solve

$$\begin{aligned} d_1r + e_1s &= pa_1 + qb_1, \\ d_2r + e_2s &= pa_2 + qb_2. \end{aligned}$$

The solution (r, s) is exactly $[\mathbf{u}]_{\mathcal{C}}$.

The basis-change procedure

A change from one basis to another consists of two familiar steps:

$$\text{old coordinates} \longrightarrow \text{the geometric vector} \longrightarrow \text{new coordinates}.$$

At no stage does the geometric vector itself change.

2.2.6 Why Perpendicular Projection Is Special

Rectangular and oblique coordinates

For the Cartesian basis, the coordinate directions are perpendicular unit vectors, so coordinates can be read from perpendicular projections. For a slanted basis this is generally false. Perpendicular shadows on $\mathbf{b}_1$ and $\mathbf{b}_2$ need not add back to $\mathbf{u}$.

The correct geometric construction draws through the endpoint lines parallel to the other basis vector. The resulting oblique parallelogram records the basis coefficients.

Remark

After matrices are introduced, the same systems of equations will be written more compactly. Matrix notation will not create a new method; it will abbreviate the reconstruction and elimination calculations performed here.

2.2.7 A Negative Coordinate Means Motion in the Opposite Direction

Example

The example $[\mathbf{u}]_{\mathcal{B}} = (2, 1)$ used positive coefficients. That is convenient for drawing a parallelogram, but basis coordinates need not be positive.

Keep

$$\mathbf{b}_1 = (2, 1), \quad \mathbf{b}_2 = (1, 2),$$

and consider

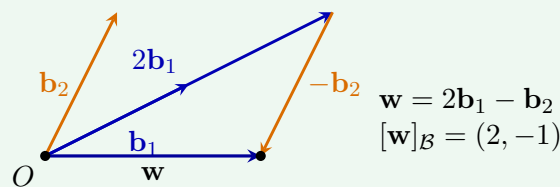
$$\mathbf{w} = 2\mathbf{b}_1 - \mathbf{b}_2.$$

Calculating with ordinary coordinate pairs gives

$$\mathbf{w} = 2(2, 1) - (1, 2) = (4, 2) - (1, 2) = (3, 0).$$

Therefore

$$[\mathbf{w}]_{\mathcal{E}} = (3, 0), \quad [\mathbf{w}]_{\mathcal{B}} = (2, -1).$$



The coefficient -1 does not mean a negative length. It means one copy of $\mathbf{b}_2$ taken in the opposite direction. Basis coordinates are signed instructions for reconstructing the vector.

Basis coordinates

A basis is an ordered pair of nonparallel vectors. Every vector in the plane has one coordinate pair relative to that basis. Positive and negative coefficients are signed instructions telling us how much of each basis vector is required. Different bases give different coordinate pairs for the same geometric vector. To change coordinates, reconstruct the vector from the old basis and then solve for its coefficients in the new basis.

2.3 Beyond a Change of Basis: Polar Coordinates

The preceding section changed the basis of the plane while keeping two fixed, nonparallel vector directions. Polar coordinates make a more radical change: they describe a point by distance and direction rather than by coefficients of two fixed basis vectors.

Linear and nonlinear coordinate changes

A basis change keeps two fixed straight directions and produces a grid of parallel lines. Polar coordinates use rays and concentric circles, and the direction used to read the coordinates varies from point to point.

Two Different Questions About the Same Point

The polar questions

Instead of asking for two signed projections, ask:

1. How far is the point from the origin?
2. In which direction must we move from the origin to reach it?

Definition

For a point different from the origin, its polar coordinates are

$$(r, \theta), \quad r > 0, \quad 0 \leq \theta < 2\pi,$$

where r is the Euclidean distance from the origin and θ is measured counterclockwise from the positive x -axis.

Definition

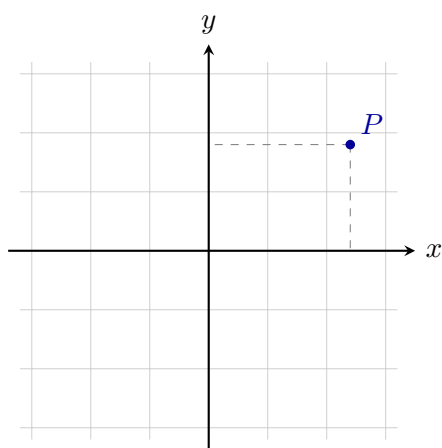
To make the representation unique in this course, the origin is represented only by $(0, 0)$. The allowed polar coordinate set is

$$\mathcal{P} = \{(0, 0)\} \cup ((0, \infty) \times [0, 2\pi)).$$

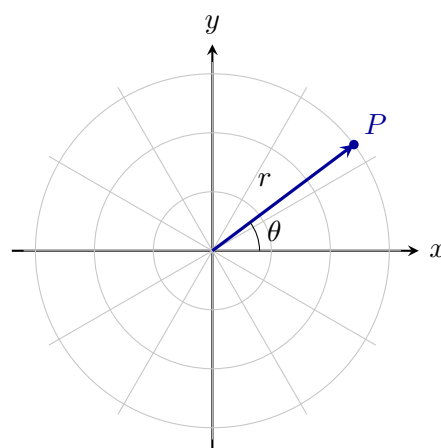
Remark

In the usual nonunique convention, every pair $(0, \theta)$ represents the origin. The restricted convention above is chosen only to give each point one allowed coordinate pair.

Rectangular and Polar Grids



Cartesian grid
two families of parallel lines



polar grid
rays and concentric circles

Coordinate curves

In Cartesian coordinates, $x = \text{constant}$ and $y = \text{constant}$ are straight lines. In polar coordinates, $\theta = \text{constant}$ is a ray and $r = \text{constant}$ is a circle. This visible difference shows that polar coordinates are not merely a slanted basis.

Converting Between the Descriptions

From polar to Cartesian

The right triangle determined by a point with polar coordinates (r, θ) gives

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Hence

$$\overrightarrow{OP} = r(\cos \theta, \sin \theta).$$

Theorem

The conversion from polar to Cartesian coordinates is

$$x = r \cos \theta, \quad y = r \sin \theta.$$

From Cartesian to polar

For $(x, y) \neq (0, 0)$, the Euclidean distance formula gives

$$r = \sqrt{x^2 + y^2}.$$

The direction is the unique $\theta \in [0, 2\pi)$ satisfying

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}.$$

Theorem

For $(x, y) \neq (0, 0)$,

$$r = \sqrt{x^2 + y^2},$$

and θ is determined by

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}.$$

Remark

The expression $\arctan(y/x)$ is insufficient because it loses quadrant information and is undefined when $x = 0$. In computation, the two-argument function $\text{atan2}(y, x)$ is used, with a negative output shifted by 2π when the chosen range is $[0, 2\pi)$.

Example

For

$$(r, \theta) = \left(2, \frac{\pi}{3}\right),$$

we obtain

$$x = 2 \cos \frac{\pi}{3} = 1, \quad y = 2 \sin \frac{\pi}{3} = \sqrt{3}.$$

Thus the same point is written as $P = (1, \sqrt{3})$ in Cartesian coordinates and as $P = (2, \pi/3)$ in polar coordinates.

Why the Transformation Is Nonlinear

Example

The polar pairs

$$(1, 0) \quad \text{and} \quad \left(1, \frac{\pi}{2}\right)$$

represent Cartesian vectors $(1, 0)$ and $(0, 1)$. Their sum is $(1, 1)$, whose polar coordinates are

$$\left(\sqrt{2}, \frac{\pi}{4}\right),$$

not $(2, \pi/2)$. Polar coordinate pairs therefore cannot be added componentwise to add vectors.

What nonlinearity means here

The formulas

$$x = r \cos \theta, \quad y = r \sin \theta$$

contain trigonometric functions. The direction $(\cos \theta, \sin \theta)$ changes with θ , so r and θ are not coefficients of two fixed basis vectors.

A Circle Becomes Simple

Let

$$C_R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}, \quad R > 0.$$

Rewriting the circle

Substitute

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = R^2.$$

Using $\cos^2 \theta + \sin^2 \theta = 1$ gives

$$r^2 = R^2.$$

Since $r \geq 0$ and $R > 0$, this reduces to $r = R$.

Theorem

The Cartesian equation

$$x^2 + y^2 = R^2$$

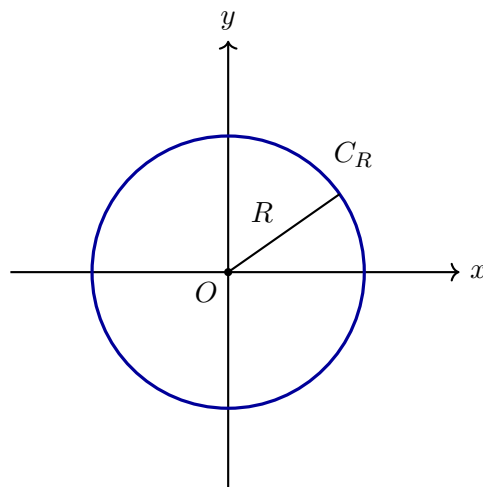
and the polar equation

$$r = R$$

describe the same circle.

Choose coordinates adapted to the geometry

Polar coordinates are efficient when distance from the origin is the dominant feature. The geometric set does not change; only its coordinate language becomes simpler.



First nonlinear coordinate system

A basis uses fixed directions and leads to linear coordinate changes. Polar coordinates use a distance and an angle:

$$(r, \theta) \longleftrightarrow (r \cos \theta, r \sin \theta).$$

Their coordinate curves are circles and rays, and vector addition is not performed by adding polar coordinate pairs.

Chapter 3

Lines and Planes in the Plane and in Space

The organising question

Lines and planes are geometric sets generated by allowed directions of motion. One point and one nonzero direction generate a line; one point and two independent directions generate a plane. Parametric, normal, implicit, and slope forms are different descriptions of those same sets, chosen to answer different questions.

3.1 Lines in the Cartesian Plane

A line should not first appear as a formula to be recognised. We already have all the ingredients needed to construct it geometrically: a point tells us where to begin, and a nonzero vector tells us in which direction motion is allowed. The question is therefore:

Which points can be reached from one fixed point by moving only in one fixed direction?

Data that determine a line

A line in the Cartesian plane is determined by one point P_0 on the line and one nonzero direction vector $\mathbf{v}$. Every point of the line is reached by adding a scalar multiple of $\mathbf{v}$ to P_0 .

Let

$$P_0 = (x_0, y_0)$$

be fixed and let

$$\mathbf{v} = (u, v) \neq \mathbf{0}$$

be a fixed direction vector. Every permitted displacement is a scalar multiple $t\mathbf{v}$. Starting from P_0 gives

$$P = P_0 + t\mathbf{v}, \quad t \in \mathbb{R}.$$

Definition

The line through P_0 with direction vector $\mathbf{v} \neq \mathbf{0}$ is

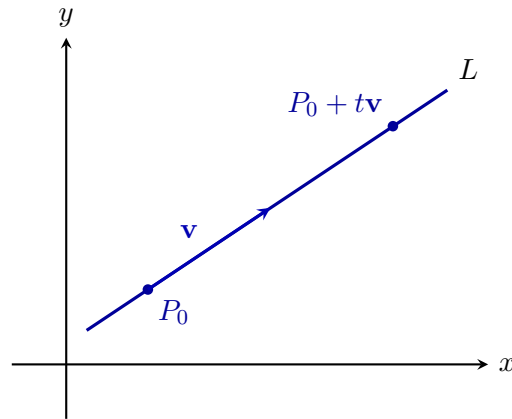
$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{aligned} x &= x_0 + tu, \\ y &= y_0 + tv, \end{aligned} \quad t \in \mathbb{R}.$$

Meaning of the parameter

At $t = 0$ we are at P_0 ; positive and negative values move in the two opposite orientations along the same direction. Changing the length of $\mathbf{v}$ only changes how quickly the parameter runs through the points, not the geometric line itself.



3.1.1 The same line described by a perpendicular direction

A direction vector tells us how to move along the line. There is a second, equally important description: choose a nonzero vector $\mathbf{n} = (a, b)$ perpendicular to $\mathbf{v}$. A point $P = (x, y)$ lies on the line precisely when the displacement $P - P_0$ is perpendicular to $\mathbf{n}$.

From direction to normal form

The dot-product test from the previous chapter gives

$$\mathbf{n} \cdot (P - P_0) = 0.$$

Written in coordinates,

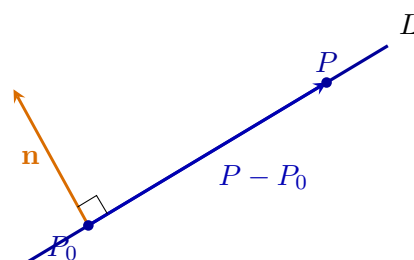
$$a(x - x_0) + b(y - y_0) = 0.$$

After collecting the constant terms,

$$ax + by = c, \quad c = ax_0 + by_0.$$

Interpretation

The general linear equation is not a second, unrelated definition of a line. It records the same object by specifying a direction that no displacement along the line may contain.



3.1.2 Slope form as a special coordinate description

Eliminating the parameter

If the horizontal component u of the direction vector is nonzero, eliminate t from

$$x = x_0 + tu, \quad y = y_0 + tv.$$

Since

$$t = \frac{x - x_0}{u},$$

we obtain

$$y - y_0 = \frac{v}{u}(x - x_0).$$

Therefore the slope is

$$m = \frac{v}{u},$$

and after collecting constants the equation takes the familiar form

$$y = mx + b.$$

Remark

If $u = 0$, the line is vertical and is described by $x = x_0$. The vector and normal descriptions treat this case without any exception; only slope notation fails.

Summary

The three forms answer different questions about the same set:

$P_0 + t\mathbf{v}$		how to move along the line,
$\mathbf{n} \cdot (P - P_0) = 0$		which displacements are allowed,
$y = mx + b$		how vertical position depends on horizontal position.

No representation is the line itself. The line is the geometric set of points that all these descriptions identify.

3.1.3 What the Familiar Cartesian Forms Let Us Read

Choose the representation adapted to the question

The parametric, normal, and slope forms are not competing definitions. They expose different information about the same geometric line, so the useful form depends on what must be calculated or interpreted.

Intersections with the coordinate axes

Impose both defining conditions

For a nonvertical line

$$y = mx + b,$$

setting $x = 0$ imposes the condition for the y -axis and gives $(0, b)$. If $m \neq 0$, setting $y = 0$ imposes the condition for the x -axis and gives

$$x = -\frac{b}{m}.$$

Theorem

For $y = mx + b$,

$$L \cap \{x = 0\} = \{(0, b)\},$$

and, when $m \neq 0$,

$$L \cap \{y = 0\} = \left\{ \left(-\frac{b}{m}, 0 \right) \right\}.$$

General intersection principle

To find an intersection of two sets, impose both defining conditions at once. Axis intercepts are the simplest example of this general method.

Example

For

$$y = -\frac{1}{2}x + 3,$$

the y -axis intersection is $(0, 3)$. Setting $y = 0$ gives $x = 6$, so the x -axis intersection is $(6, 0)$.

The line through two points

Extracting a direction from two points

Let

$$P = (x_1, y_1), \quad Q = (x_2, y_2), \quad P \neq Q.$$

The displacement

$$Q - P = (x_2 - x_1, y_2 - y_1)$$

is nonzero and therefore provides a direction vector of the unique line through the points.

Theorem

The unique line through distinct points P and Q is

$$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$

This formula handles vertical and nonvertical lines uniformly.

Recovering slope notation

If $x_1 \neq x_2$, the horizontal component of $Q - P$ is nonzero, so

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

and

$$y - y_1 = m(x - x_1).$$

If $x_1 = x_2$, the line is vertical and is described by

$$x = x_1.$$

Remark

The exceptional vertical case belongs only to slope notation. The geometric line and its parametric description require no exception.

Parallel lines in slope notation

Why equal slopes mean equal directions

The nonvertical lines

$$y = m_1x + b_1, \quad y = m_2x + b_2$$

have direction vectors $(1, m_1)$ and $(1, m_2)$. These directions are parallel exactly when $m_1 = m_2$.

Theorem

For distinct nonvertical lines,

$$L_1 \parallel L_2 \iff m_1 = m_2 \text{ and } b_1 \neq b_2.$$

If also $b_1 = b_2$, the two equations describe the same line.

Remark

Slope formulas are efficient coordinate consequences of the vector construction. They should be used when they simplify a question, but not at the cost of hiding the more general descriptions that handle vertical lines and extend directly to space.

3.1.4 Vector Tests Behind the Slope Rules

Theorem

Let two lines have nonzero direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Their directions are parallel precisely when

$$\mathbf{v}_1 \parallel \mathbf{v}_2 \iff \mathbf{v}_2 = \lambda \mathbf{v}_1 \text{ for some } \lambda \neq 0.$$

They are perpendicular precisely when

$$\mathbf{v}_1 \perp \mathbf{v}_2 \iff \mathbf{v}_1 \cdot \mathbf{v}_2 = 0.$$

Why vector tests are more general

These criteria do not require a slope and therefore include vertical directions without any special convention. In three dimensions, parallel directions will also be detected by a zero cross product.

Recovering the familiar slope rule

For nonvertical lines with slopes m_1, m_2 , choose direction vectors

$$\mathbf{v}_1 = (1, m_1), \quad \mathbf{v}_2 = (1, m_2).$$

Then

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 + m_1 m_2.$$

Therefore

$$\mathbf{v}_1 \perp \mathbf{v}_2 \iff 1 + m_1 m_2 = 0.$$

Theorem

For two nonvertical lines,

$$L_1 \perp L_2 \iff m_1 m_2 = -1.$$

This is the slope form of the dot-product criterion, not an independent rule.

3.1.5 The Nearest Point on a Line

Let

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}, \quad \mathbf{v} \neq \mathbf{0},$$

and let P be any point.

Projecting onto the allowed direction

Decompose the displacement $P - P_0$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$. The parallel component is

$$\text{proj}_{\mathbf{v}}(P - P_0) = \frac{(P - P_0) \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Starting from P_0 and moving by this component reaches

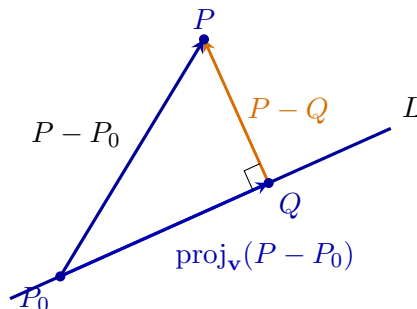
$$Q = P_0 + \text{proj}_{\mathbf{v}}(P - P_0).$$

Theorem

The nearest point on L to P is

$$Q = P_0 + \text{proj}_{\mathbf{v}}(P - P_0).$$

The remainder $P - Q$ is perpendicular to the line.



Definition

The distance from P to the line L is

$$d(P, L) = \|(P - P_0) - \text{proj}_{\mathbf{v}}(P - P_0)\|.$$

Why the perpendicular is shortest

Every other route from P to a point of the line contains the same perpendicular component together with an additional component along the line. The orthogonal component alone is therefore the shortest possible displacement.

3.1.6 Distance from the Normal Equation

Suppose the line is written as

$$ax + by = c, \quad (a, b) \neq (0, 0),$$

with normal vector $\mathbf{n} = (a, b)$.

Reading the normal component

For $P = (x_P, y_P)$ and any fixed P_0 on the line,

$$\mathbf{n} \cdot (P - P_0) = ax_P + by_P - c.$$

The signed scalar component in the normal direction is

$$\frac{ax_P + by_P - c}{\sqrt{a^2 + b^2}}.$$

Distance is its absolute value.

Theorem

For the line $ax + by = c$ and the point $P = (x_P, y_P)$,

$$d(P, L) = \frac{|ax_P + by_P - c|}{\sqrt{a^2 + b^2}}.$$

The formula is a projection

The numerator measures signed displacement in the normal direction, and the denominator normalises the normal vector. The formula is therefore the length of a projection, not a detached rule to memorise.

3.2 Lines in Three-Dimensional Space

The same construction in one more coordinate

A line in space is still determined by one point and one nonzero direction. The third coordinate changes the arithmetic, but not the geometric idea.

Definition

Let

$$P_0 = (x_0, y_0, z_0), \quad \mathbf{v} = (a, b, c) \neq \mathbf{0}.$$

The line through P_0 with direction $\mathbf{v}$ is

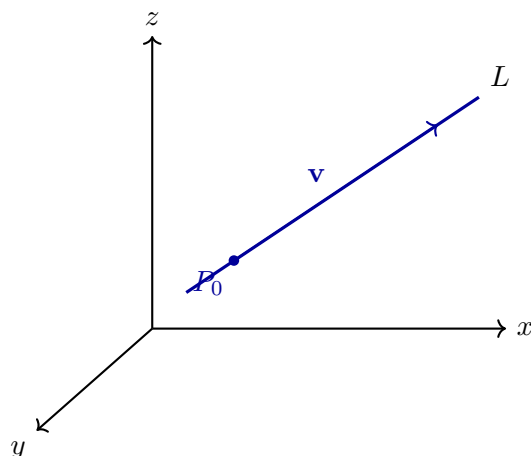
$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

Equivalently,

$$\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \\ z = z_0 + ct. \end{cases}$$

Meaning of the parameter

As t varies, the point moves along the line. Positive and negative values give the two orientations. Rescaling the direction vector changes only the speed at which the parameter traverses the same set of points.



3.2.1 The Line Through Two Points in Space

The direction is the displacement between the points

Let

$$P = (x_1, y_1, z_1), \quad Q = (x_2, y_2, z_2)$$

be distinct. Then

$$Q - P = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

is a nonzero direction vector.

Theorem

The unique line through distinct points $P, Q \in \mathbb{R}^3$ is

$$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{cases} x = x_1 + t(x_2 - x_1), \\ y = y_1 + t(y_2 - y_1), \\ z = z_1 + t(z_2 - z_1). \end{cases}$$

Example

For

$$P = (1, 0, 2), \quad Q = (3, 4, 1),$$

a direction vector is $Q - P = (2, 4, -1)$. Therefore

$$(x, y, z) = (1, 0, 2) + t(2, 4, -1),$$

or

$$\begin{cases} x = 1 + 2t, \\ y = 4t, \\ z = 2 - t. \end{cases}$$

3.2.2 Symmetric Form

Eliminating the parameter

If a, b, c are all nonzero, then

$$t = \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Hence

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Remark

If a direction component is zero, the corresponding coordinate is constant. For example, $c = 0$ gives $z = z_0$. The parametric form handles this without division by zero and is therefore the more fundamental representation.

3.2.3 Parallel, Perpendicular, Intersecting, and Skew Lines

Consider

$$L_1 = P_1 + t\mathbf{v}_1, \quad L_2 = P_2 + s\mathbf{v}_2.$$

Theorem

The directions are parallel precisely when

$$\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{0}.$$

They are perpendicular precisely when

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0.$$

Remark

Parallel directions do not by themselves distinguish identical lines from distinct parallel lines; one must also test whether $P_2 - P_1$ is parallel to the common direction. Likewise, perpendicular directions do not make the geometric lines perpendicular unless the lines also intersect.

Testing intersection

Solve

$$P_1 + t\mathbf{v}_1 = P_2 + s\mathbf{v}_2.$$

This gives three scalar equations for two parameters. A common solution produces the intersection point. If the directions are nonparallel but the system is inconsistent, the lines are skew.

Definition

Two lines in $\mathbb{R}^3$ are *skew* when they are neither parallel nor intersecting.

Example

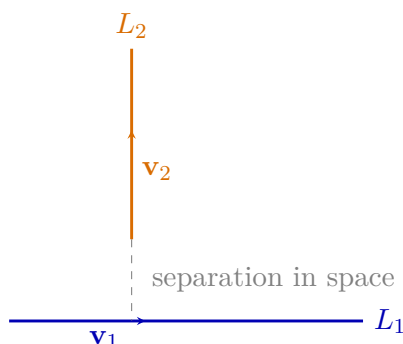
Consider

$$L_1 = (0, 0, 0) + t(1, 0, 0)$$

and

$$L_2 = (0, 1, 1) + s(0, 1, 0).$$

Their direction vectors have dot product zero. Yet every point on L_1 has third coordinate 0, while every point on L_2 has third coordinate 1. Thus the lines do not intersect: their directions are perpendicular, but the lines are skew.



A genuinely three-dimensional possibility

Skew lines have no analogue in the plane. In two dimensions, distinct nonparallel lines must intersect. In space, one line can pass above or beside another without meeting it.

Lines in space

A line is $P_0 + t\mathbf{v}$. Two points determine the direction $Q - P$. A zero cross product detects parallel directions, a zero dot product detects perpendicular directions, and a parameter system distinguishes intersection from skewness.

3.3 Planes in Three-Dimensional Space

A line was produced by allowing one real parameter:

$$P = P_0 + t\mathbf{v}.$$

Only one independent direction of motion was available.

The geometric question

What set is obtained when motion from one fixed point is allowed in two independent directions?

Let $P_0 \in \mathbb{R}^3$ be fixed and let $\mathbf{u}, \mathbf{v}$ be nonparallel vectors.

Generating the plane

Every permitted displacement is a linear combination

$$s\mathbf{u} + t\mathbf{v}.$$

Starting at P_0 gives

$$P = P_0 + s\mathbf{u} + t\mathbf{v}.$$

One parameter produces a line; two independent parameters produce a plane. If the two directions were parallel, both parameters would still move only along one line.

Definition

The plane through P_0 generated by nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ is

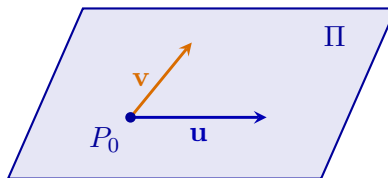
$$\Pi = \{P_0 + s\mathbf{u} + t\mathbf{v} : s, t \in \mathbb{R}\}.$$

If

$$P_0 = (x_0, y_0, z_0), \quad \mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

then

$$\begin{cases} x = x_0 + su_1 + tv_1, \\ y = y_0 + su_2 + tv_2, \\ z = z_0 + su_3 + tv_3. \end{cases}$$



3.3.1 Why a Normal Vector Appears

The cross product is perpendicular to every allowed motion

The vector chapter proved that

$$(\mathbf{u} \times \mathbf{v}) \cdot (a\mathbf{u} + b\mathbf{v}) = 0$$

for all real a, b . Therefore

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}$$

is perpendicular to every displacement allowed in the plane.

From the normal direction to an equation

A point P belongs to the plane exactly when $P - P_0$ is an allowed combination. Hence

$$\mathbf{n} \cdot (P - P_0) = 0.$$

For $\mathbf{n} = (a, b, c)$ this becomes

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or, after collecting constants,

$$ax + by + cz = d.$$

Theorem

The parametric and normal descriptions of the same plane are

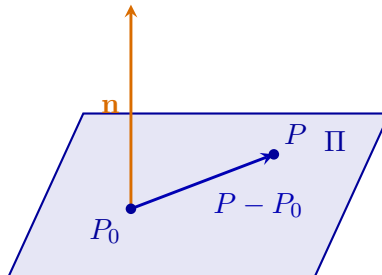
$$P = P_0 + s\mathbf{u} + t\mathbf{v}$$

and

$$\mathbf{n} \cdot (P - P_0) = 0 \iff ax + by + cz = d.$$

Two descriptions, one plane

The parametric equation lists all permitted motions in the plane. The normal equation states that the forbidden normal component of every in-plane displacement must vanish.



3.3.2 A Plane Through Three Points

Extracting two independent directions

Let P, Q, R be three noncollinear points. Then

$$\mathbf{u} = Q - P, \quad \mathbf{v} = R - P$$

are nonparallel and provide the two directions required to generate a plane. Their cross product

$$\mathbf{n} = (Q - P) \times (R - P)$$

is nonzero and perpendicular to the plane.

Theorem

The unique plane through three noncollinear points P, Q, R is

$$\Pi = \{P + s(Q - P) + t(R - P) : s, t \in \mathbb{R}\}.$$

A normal vector is

$$\mathbf{n} = (Q - P) \times (R - P).$$

Example

For

$$P = (1, 0, 0), \quad Q = (0, 1, 0), \quad R = (0, 0, 1),$$

we obtain

$$Q - P = (-1, 1, 0), \quad R - P = (-1, 0, 1),$$

and

$$(Q - P) \times (R - P) = (1, 1, 1).$$

Using the point P gives

$$(1, 1, 1) \cdot ((x, y, z) - (1, 0, 0)) = 0,$$

so

$$x + y + z = 1.$$

3.3.3 Relative Positions of Planes

Let

$$\Pi_1 : \mathbf{n}_1 \cdot P = d_1, \quad \Pi_2 : \mathbf{n}_2 \cdot P = d_2.$$

Normals control relative position

If the normals are parallel, the planes have the same orientation and are either identical or distinct and parallel. If the normals are not parallel, the planes intersect in a line.

Theorem

The angle between planes is measured through their normals. In particular,

$$\Pi_1 \perp \Pi_2 \iff \mathbf{n}_1 \cdot \mathbf{n}_2 = 0.$$

If the planes are nonparallel, a direction vector of their intersection line is

$$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2.$$

Finding the intersection line

The cross product gives the direction because the line must be perpendicular to both normals. A point on the line is then found by solving the two plane equations simultaneously.

3.3.4 Distance from a Point to a Plane

Distance as a normal projection

Let P_0 be a fixed point on the plane. The shortest displacement from P to the plane follows the normal direction. Hence the distance is the magnitude of the scalar component of $P - P_0$ along $\mathbf{n}$:

$$d(P, \Pi) = \frac{|\mathbf{n} \cdot (P - P_0)|}{\|\mathbf{n}\|}.$$

Theorem

For a plane $ax + by + cz = d$ and a point $P = (x_P, y_P, z_P)$,

$$d(P, \Pi) = \frac{|ax_P + by_P + cz_P - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

What the formula measures

The numerator is the signed normal displacement encoded by the plane equation; the denominator converts the chosen normal vector to unit length. The formula is an orthogonal projection, not an isolated computational rule.

Line and plane from permitted directions

A line is generated by one independent direction,

$$P = P_0 + t\mathbf{v},$$

whereas a plane is generated by two,

$$P = P_0 + s\mathbf{u} + t\mathbf{v}.$$

The cross product of the two generating directions produces a normal, and the dot product turns perpendicularity into the equation

$$\mathbf{n} \cdot (P - P_0) = 0.$$

The parametric and normal forms therefore arise from the same geometric construction.

Chapter 4

Conic Sections and Quadric Surfaces

Geometry first, recognition second

Conic sections are introduced through geometric distance conditions and only then translated into quadratic equations. Quadric surfaces are understood by cutting them with planes and recognising the resulting conics. Equations are a language for the objects; they are not substitutes for the geometric objects.

4.1 Conic Sections: From Geometric Conditions to Equations

The previous chapter studied sets determined by first-degree equations. We now ask what curves arise when distances between moving and fixed geometric objects are constrained.

Why quadratic equations appear

Euclidean distance contains squares. When a geometric distance condition is translated into Cartesian coordinates and square roots are removed, second-degree terms arise naturally. Quadratic equations are consequences of geometry, not an arbitrary next list of formulas.

4.1.1 A Circle: One Fixed Point and One Fixed Distance

Definition

Fix a centre $C = (h, k)$ and a number $r > 0$. The circle with centre C and radius r is the set of points $P = (x, y)$ satisfying

$$d(P, C) = r.$$

Cartesian equation of the circle

Using the Euclidean distance formula,

$$\sqrt{(x - h)^2 + (y - k)^2} = r.$$

Both sides are nonnegative, so squaring preserves the condition and gives

$$(x - h)^2 + (y - k)^2 = r^2.$$

Definition versus equation

The equation is not the definition of the circle. It is the Cartesian expression of the geometric requirement that every point remain a fixed distance from one centre.

4.1.2 A Parabola: Equal Distance from a Point and a Line

Definition

Fix a point F , called the focus, and a line D , called the directrix. A parabola is the set of points P satisfying

$$d(P, F) = d(P, D).$$

Choose coordinates so that

$$F = (0, p), \quad D : y = -p, \quad p > 0.$$

Cartesian equation of the parabola

For $P = (x, y)$,

$$d(P, F) = \sqrt{x^2 + (y - p)^2}, \quad d(P, D) = |y + p|.$$

On the resulting parabola $y \geq -p$, so $|y + p| = y + p$. Hence

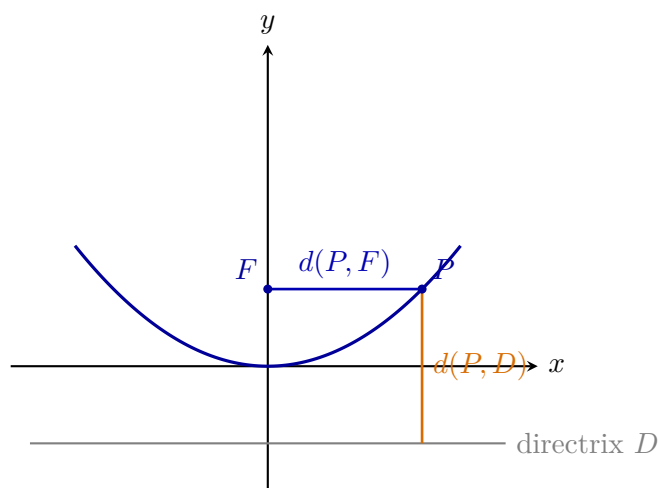
$$\sqrt{x^2 + (y - p)^2} = y + p.$$

Squaring and cancelling equal terms gives

$$x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2,$$

and therefore

$$x^2 = 4py.$$



What the parabola equation records

The equation $x^2 = 4py$ is the Cartesian form of the equal-distance condition. It should always be read together with the focus and directrix geometry that produced it.

4.1.3 An Ellipse: A Fixed Sum of Two Distances

Definition

Fix two foci

$$F_1 = (-c, 0), \quad F_2 = (c, 0), \quad c > 0,$$

and choose $a > c$. An ellipse is the set of points P satisfying

$$d(P, F_1) + d(P, F_2) = 2a.$$

Remark

The inequality $a > c$ is necessary: the prescribed sum must exceed the distance $2c$ between the foci.

Cartesian equation of the ellipse

Write

$$r_1 = \sqrt{(x+c)^2 + y^2}, \quad r_2 = \sqrt{(x-c)^2 + y^2}.$$

The condition is $r_1 + r_2 = 2a$. Isolating and squaring twice gives

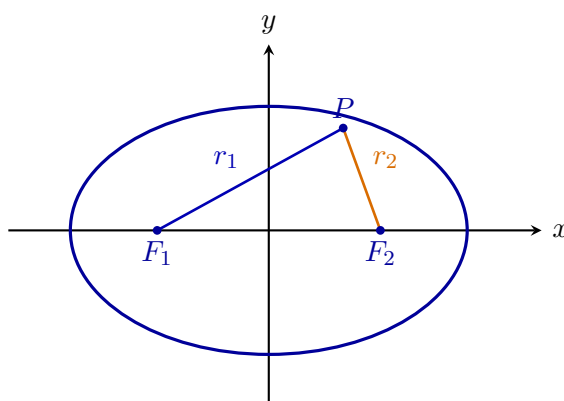
$$\left(1 - \frac{c^2}{a^2}\right) x^2 + y^2 = a^2 - c^2.$$

Define

$$b^2 = a^2 - c^2 > 0.$$

Then

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$



Why the ellipse is bounded

The total journey from one focus to the moving point and then to the other focus is fixed. This constraint prevents the point from moving arbitrarily far away and produces a bounded oval.

4.1.4 A Hyperbola: A Fixed Difference of Two Distances

Definition

With the same foci, choose $0 < a < c$. A hyperbola is the set of points P satisfying

$$|d(P, F_1) - d(P, F_2)| = 2a.$$

Cartesian equation of the hyperbola

On the right branch, write $r_1 - r_2 = 2a$. Isolating the remaining square root and squaring twice gives

$$\left(\frac{c^2}{a^2} - 1\right)x^2 - y^2 = c^2 - a^2.$$

Set

$$b^2 = c^2 - a^2 > 0.$$

Then

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Why two branches appear

The absolute value allows either focus to be nearer. Changing a fixed sum into a fixed difference changes a bounded curve into two unbounded branches. The new geometry comes from changing the distance condition, not from choosing an unrelated formula.

4.1.5 Why These Curves Form One Family

One spatial experiment

Intersect a double cone with a plane and vary the inclination of the cutting plane. A perpendicular cut gives a circle; a tilted cut meeting one nappe gives an ellipse; the limiting parallel cut gives a parabola; and a cut meeting both nappes gives a hyperbola.

The unity of conic sections

The four curves are not an arbitrary list. They arise by varying one geometric operation while keeping the cone fixed. Their different equations record different outcomes of the same spatial construction.

Remark

The rest of the chapter studies how these geometric sets appear in Cartesian equations. Standard forms, translations, rotations, and discriminants are tools for recognising the object already present in an equation. They do not replace the geometric definition of the curve.

Geometric definition before algebraic recognition

A circle fixes one distance, a parabola equates two distances, an ellipse fixes a sum, and a hyperbola fixes a difference. In each case the quadratic equation is derived from the

geometric condition and should be interpreted through it.

4.1.6 From Construction to Recognition

The geometric definitions answer the question *what the curves are*. A different problem begins when an equation is given first.

The recognition question

Which geometric curve, if any, is hidden in a given second-degree algebraic condition?

Two different tasks

Constructing a conic starts from a geometric condition and derives an equation. Recognising a conic starts from an equation and reconstructs the geometry. Standard forms, completing squares, translations, rotations, and discriminants belong to the second task; they do not replace the geometric definition.

First visual clues in the equation

When no mixed term is present, squared terms with the same sign suggest an ellipse-type set, opposite signs suggest hyperbolic behaviour, and the absence of one quadratic direction suggests a parabola. Degenerate cases may instead produce points, lines, or no real points.

What the mixed term means

A term Bxy indicates that the chosen coordinate axes may not follow the natural symmetry directions of the curve. Rotating the coordinate system can remove the mixed term and reveal the standard geometry.

Remark

This subsection motivates the recognition problem. The precise classification criterion and its exceptional cases are stated once in the algebraic classification that follows.

Algebraic Classification of Second-Degree Curves

Lines arise from equations in which the coordinates occur only to the first power. The next natural class allows terms of total degree two.

Definition

A *second-degree equation* in the Cartesian plane has the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where $A, B, C, D, E, F \in \mathbb{R}$ and at least one of A, B, C is nonzero. Its solution set consists of all points (x, y) satisfying the equation.

The role of the quadratic part

The terms

$$Ax^2 + Bxy + Cy^2$$

form the quadratic part. They largely determine the geometric type, while the linear and constant terms mainly control position and scale.

A First Classification

Definition

For

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

define the conic discriminant

$$\Delta = B^2 - 4AC.$$

Theorem

For a nondegenerate real conic,

$\Delta < 0$	ellipse type, including circles, parabola type, hyperbola type.
$\Delta = 0$	
$\Delta > 0$	

The sign is preserved by translations and rotations of the coordinate system.

Why translations and rotations are used

A translation moves the centre or vertex to a simpler position. A rotation aligns the axes with the natural symmetry directions and can remove the mixed term Bxy . These transformations simplify the description without changing the underlying geometric curve.

Remark

The sign of Δ identifies the conic type only after degenerate cases are separated. A second-degree equation may describe a point, no real points, one line, or two lines.

Circle

Definition

A circle with centre (h, k) and radius $r > 0$ has standard equation

$$(x - h)^2 + (y - k)^2 = r^2.$$

Classification by the discriminant

Expanding gives

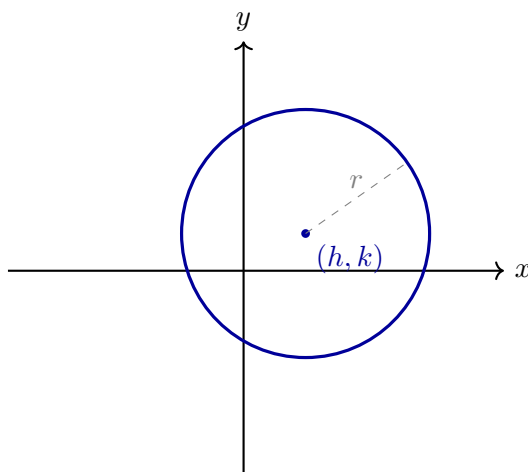
$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0.$$

Here $A = C = 1$ and $B = 0$, so

$$\Delta = -4 < 0.$$

Circle as a special ellipse

A circle is an ellipse-type conic whose two quadratic coefficients are equal and whose natural axes have the same scale.



Ellipse

Definition

A standard ellipse centred at the origin has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > 0, \quad b > 0.$$

The parameters a and b are the semi-axis lengths.

Classification by the discriminant

Multiplying by a^2b^2 gives

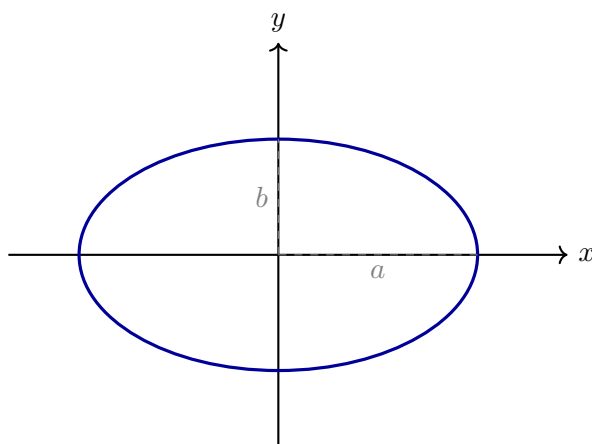
$$b^2x^2 + a^2y^2 - a^2b^2 = 0.$$

Thus $A = b^2$, $B = 0$, and $C = a^2$, so

$$\Delta = -4a^2b^2 < 0.$$

Remark

When $a = b$, the ellipse becomes a circle.



Parabola

Definition

A standard vertical parabola has equation

$$y = ax^2, \quad a \neq 0.$$

A translated form is

$$y - k = a(x - h)^2,$$

with vertex (h, k) .

Classification by the discriminant

The equation $y = ax^2$ is equivalent to

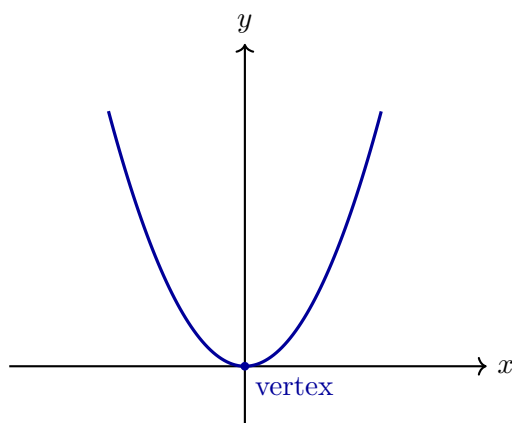
$$ax^2 - y = 0.$$

Here $A = a$, $B = 0$, and $C = 0$, so

$$\Delta = 0.$$

What the coefficient a controls

The sign of a determines whether the parabola opens upward or downward, while its magnitude controls how rapidly it widens.



Hyperbola

Definition

A standard hyperbola centred at the origin has equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad a > 0, \quad b > 0.$$

Its asymptotes are

$$y = \frac{b}{a}x, \quad y = -\frac{b}{a}x.$$

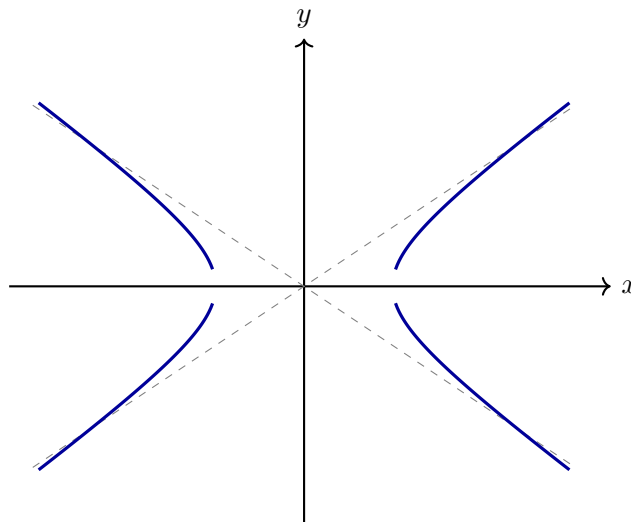
Classification by the discriminant

Multiplying by a^2b^2 gives

$$b^2x^2 - a^2y^2 - a^2b^2 = 0.$$

Thus $A = b^2$, $B = 0$, and $C = -a^2$, so

$$\Delta = 4a^2b^2 > 0.$$



The Mixed Term and Rotation

What Bxy signals

A mixed term indicates that the chosen Cartesian axes may not coincide with the natural symmetry directions of the conic.

Example

The equation

$$x^2 + 2xy + y^2 = 1$$

can be written as

$$(x + y)^2 = 1.$$

It is degenerate and describes the two parallel lines

$$x + y = 1, \quad x + y = -1.$$

Rotating coordinates

Introduce new coordinates (u, v) by

$$x = u \cos \theta - v \sin \theta, \quad y = u \sin \theta + v \cos \theta.$$

Choosing θ appropriately removes the uv term. When $A \neq C$, one may choose

$$\tan(2\theta) = \frac{B}{A - C}.$$

The resulting equation can then be classified in the rotated coordinates using standard forms.

Degenerate Second-Degree Equations

A quadratic equation need not be a conic curve

Factorisation or completing squares may reveal a point, no real points, one line, or two lines instead of a nondegenerate circle, ellipse, parabola, or hyperbola.

Example

The equation

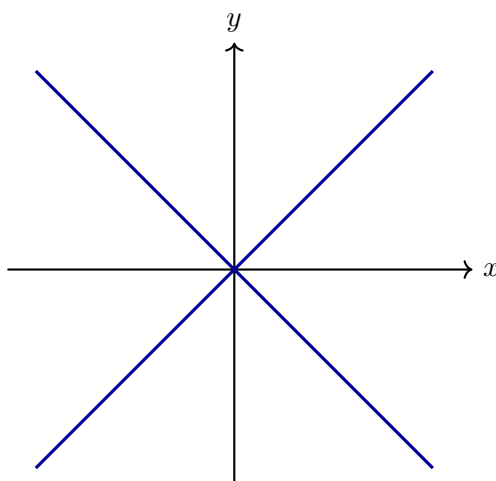
$$x^2 - y^2 = 0$$

factors as

$$(x - y)(x + y) = 0.$$

Its solution set is the union of two intersecting lines:

$$y = x \quad \text{or} \quad y = -x.$$



Example

Other common degenerate cases are

$$x^2 + y^2 = 0,$$

which gives only $(0, 0)$,

$$x^2 + y^2 + 1 = 0,$$

which has no real solutions, and

$$x^2 = 0,$$

which describes the line $x = 0$ counted algebraically twice.

Recognition procedure

Second-degree equations have the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

For nondegenerate real conics, $\Delta = B^2 - 4AC$ separates ellipse, parabola, and hyperbola types. Translations and rotations reduce equations to standard forms, while factorisation and completing squares expose degenerate cases.

4.2 Understanding Quadric Surfaces Through Sections

The method

A three-dimensional surface should not be recognised only from a memorised formula. Intersect it with simple planes, identify the resulting planar curves, and study how those curves change as the cutting plane moves.

Fixing $z = k$ gives a horizontal section; fixing $x = k$ or $y = k$ gives vertical sections. The same questions will be asked repeatedly:

1. Which curve appears in the cutting plane?
2. For which parameter values does it exist?
3. How does its size or type change?
4. What does the family of sections reveal about the whole surface?

4.2.1 Ellipsoid

Definition

An ellipsoid has standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Horizontal sections

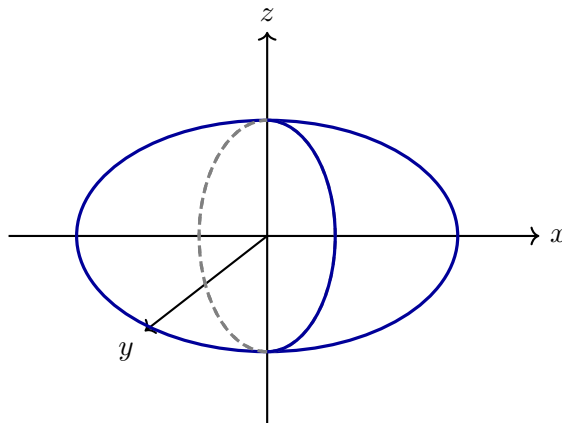
Setting $z = k$ gives

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 - \frac{k^2}{c^2}.$$

Thus $|k| < c$ gives an ellipse, $|k| = c$ gives one point, and $|k| > c$ gives no real points.

Reading the whole surface

The sections grow from one endpoint, reach their largest size at $z = 0$, and shrink to the other endpoint. The disappearance of sections beyond $|k| = c$ shows that the surface is bounded.



4.2.2 Elliptic Paraboloid

Definition

An elliptic paraboloid has standard equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}.$$

Horizontal and vertical sections

For $z = k$,

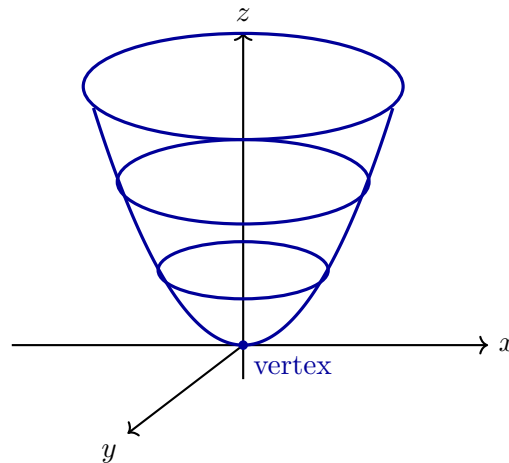
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = k.$$

There are no points for $k < 0$, one vertex for $k = 0$, and growing ellipses for $k > 0$. Setting $y = 0$ gives the parabola

$$z = \frac{x^2}{a^2}.$$

Reading the opening

Horizontal sections show that the surface widens as height increases. Vertical sections show the parabolic profile and the direction in which the surface opens.



4.2.3 Hyperbolic Paraboloid

Definition

A hyperbolic paraboloid has standard equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Sections change type and orientation

For $z = k$,

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = k.$$

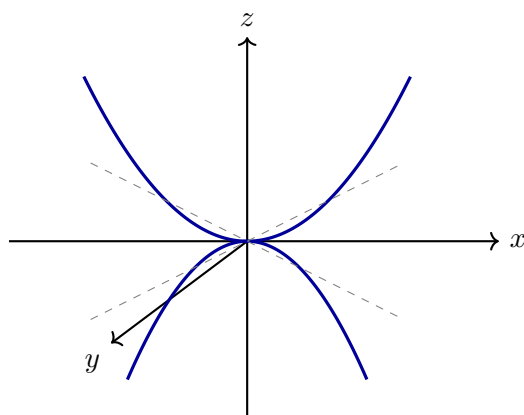
For $k > 0$ the hyperbolas open in the x direction; for $k < 0$ they open in the y direction. At $k = 0$,

$$\left(\frac{x}{a} - \frac{y}{b}\right) \left(\frac{x}{a} + \frac{y}{b}\right) = 0,$$

so the section is two intersecting lines. The vertical sections $y = 0$ and $x = 0$ are parabolas opening in opposite directions.

Why the surface is a saddle

The change of hyperbola orientation across $k = 0$, together with vertical parabolas opening oppositely, reveals a surface that bends upward in one direction and downward in the perpendicular direction.



4.2.4 Hyperboloid of One Sheet

Definition

A hyperboloid of one sheet has standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

Horizontal sections

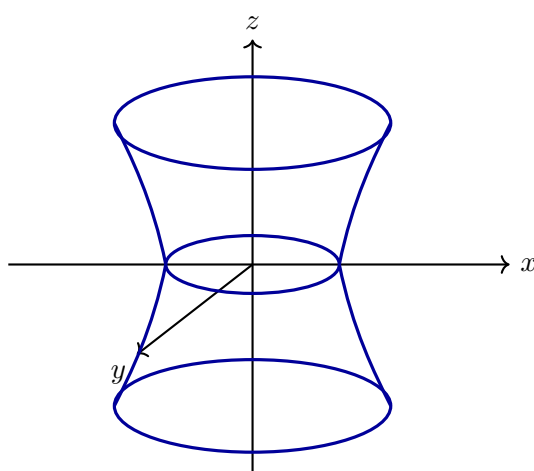
For $z = k$,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 + \frac{k^2}{c^2}.$$

Every height gives an ellipse, smallest at $k = 0$ and larger as $|k|$ grows.

Connected surface with a waist

Because a nonempty section exists at every height, the surface remains connected. The smallest central ellipse produces the visible waist.



4.2.5 Hyperboloid of Two Sheets

Definition

A hyperboloid of two sheets has standard equation

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Horizontal sections

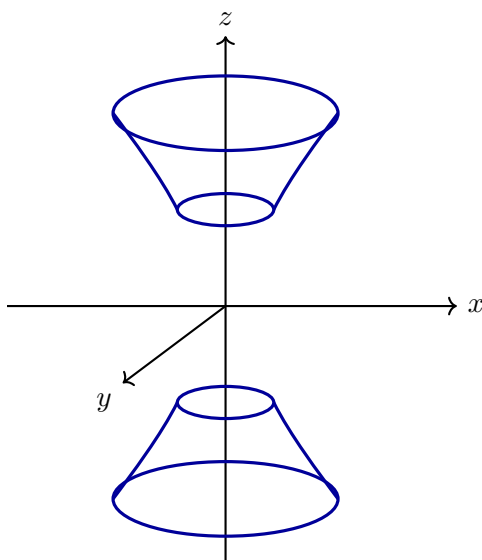
For $z = k$,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2} - 1.$$

There are no points when $|k| < c$, one point when $|k| = c$, and ellipses when $|k| > c$.

Why the surface separates

The empty band around $z = 0$ creates a genuine gap. The absence of sections in that interval explains why the surface has two disconnected components.



4.2.6 Elliptic Cone

Definition

An elliptic cone has standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

Horizontal sections

For $z = k$,

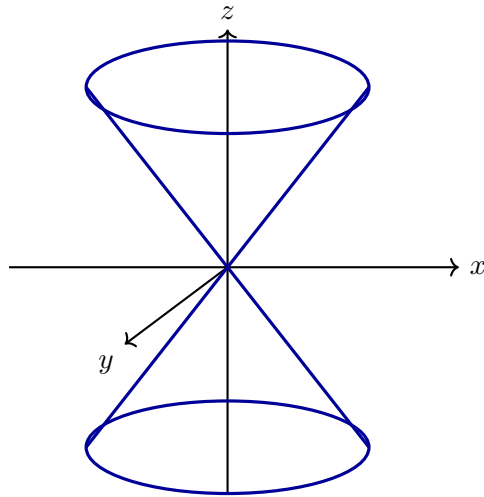
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2}.$$

At $k = 0$ the section is one point. Equal positive and negative heights give ellipses of the

same size.

Two nappes and one vertex

The symmetry in $|k|$ produces two congruent nappes meeting at the origin. The single-point section at $k = 0$ identifies the vertex.



4.2.7 Cylinders

An absent coordinate creates translation symmetry

If one coordinate does not appear in the equation, moving in that coordinate direction never changes whether the equation is satisfied. A planar curve is therefore carried unchanged along a line to produce a cylinder.

Example

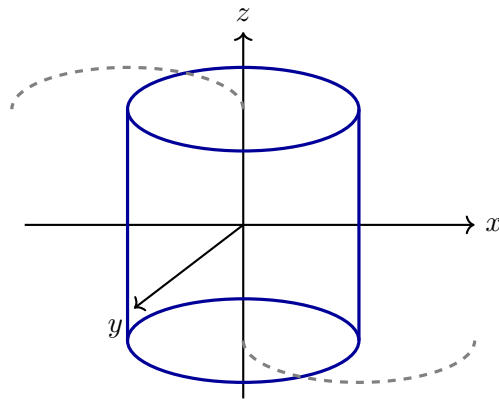
The equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

has the same elliptical section for every z and produces an elliptic cylinder. The equation

$$y = x^2$$

produces the same parabola in every plane $z = k$ and therefore gives a parabolic cylinder.



What sections reveal

A standard equation names a surface, but its sections explain it. Empty sections reveal gaps, shrinking sections reveal endpoints or vertices, changing conic types reveal saddle behaviour, and an absent coordinate reveals a direction of translation.

From curves to surfaces

A circle, parabola, ellipse, or hyperbola is first determined by a geometric condition and then represented by an equation. Completing squares and rotating coordinates recover that geometry from a general quadratic equation. In three dimensions, families of planar sections reveal whether a quadric is bounded, connected, separated, cylindrical, conical, or saddle-shaped.

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Chapter 6

Chapter 7

Chapter 8

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Chapter 12